LOCAL AND GLOBAL NULL CONTROLLABILITY FOR NONLINEAR STOCHASTIC BIHARMONIC EQUATIONS IN HIGH DIMENSIONS

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Abstract. In this paper, we study null controllability for semilinear stochastic biharmonic equations in dimensions $n \geq 2$ with Navier or clamped boundary conditions. Under an energy coercivity condition on the diffusion, we establish global null controllability with an interior drift control and a full-domain diffusion control for progressively measurable drifts vanishing at the origin and globally Lipschitz in the state, gradient and Hessian. This result resolves the higher-dimensional extension left open in [33, Remark 1.4]. In particular, if the drift is positively homogeneous of degree one and the diffusion takes the form $G(t, y) = b(t, y)y$, where b is a bounded scalar-valued function, we show that only a drift control is sufficient. Furthermore, when the drift coefficient is superlinear and satisfies dimension-dependent polynomial increment bounds, we establish a local null controllability result for small initial state. A superlinear counterexample using any interior ball outside the drift control region yields an upper bound on the existence time for every square-integrable diffusion control.

1. Introduction

Fourth-order parabolic equations arise when interfacial or bending energies are coupled with dissipative evolution. In phase separation, a gradient penalty in the free energy contributes a second-order term to the chemical potential; conservation of mass then leads to fourth-order diffusion. The curvature-dependent pressure induced by surface tension leads, through the capillary flux, to a fourth-order evolution equation in thin-film models. Curvature-dependent bending energies provide a further source of fourth-order operators in models of elastic interfaces [4, 9, 11]. Nonlinear derivative terms thus arise naturally alongside fourth-order diffusion. Thermal fluctuations also enter phase-separation and thin-film dynamics and may change the growth of perturbations and the time scale of film rupture [5, 9]. We are thus led to stochastic fourth-order equations with nonlinear drift and diffusion, whose controllability requires estimates compatible with both spatial derivatives and adaptedness.

Let $\mathcal{O} \subset \mathbb{R}^n$, $n \geq 2$, be a bounded connected domain with smooth boundary, and let $\mathcal{O}_0 \subset \mathcal{O}$ be a nonempty open control region. For any $T > 0$, set $Q = (0, T) \times \mathcal{O}$, $Q_0 = (0, T) \times \mathcal{O}_0$ and $\Sigma = (0, T) \times \partial\mathcal{O}$. We consider the following controlled semilinear stochastic biharmonic

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problem:

$$
\begin{cases}
\mathrm{d}y + \Delta^2 y \,\mathrm{d}t = \big(F(y, \nabla y, D^2 y) + \mathbf{1}_{\mathcal{O}_0} u\big)\,\mathrm{d}t + \big(g(y, \nabla y, D^2 y) + v\big)\,\mathrm{d}W_t, & \text{in } Q, \\
\mathcal{B}y = \begin{cases} (y, \Delta y), & \text{Navier}, \\ (y, \partial_\nu y), & \text{clamped} \end{cases} = (0,0), & \text{on } \Sigma, \\
y(0) = y_0, & \text{in } \mathcal{O}.
\end{cases} \tag{1.1}
$$

Here $D^2 y = (\partial_{ij} y)_{i,j=1}^n$; matrix norms are Frobenius norms. The symbol ν denotes the outward unit normal to $\partial\mathcal{O}$, and $\mathbf{1}_{\mathcal{O}_0}$ is the indicator of $\mathcal{O}_0$. The dependence of F and g on (ω, t, x) is omitted when unambiguous. The process W is a standard real-valued Brownian motion on a fixed stochastic basis; stochastic integrals are in the Itô sense.

Our main contribution is to establish global null controllability for equations whose drift depends on the state, gradient and full Hessian in several space dimensions. Under the Lipschitz and diffusion coercivity assumptions below, an interior drift control and a full-domain diffusion control drive every square-integrable initial state to zero almost surely at any prescribed positive time, for both Navier and clamped boundary conditions. We also obtain a globally Lipschitz selection of the state and controls, together with a quantitative cost bound. Our first main result, Theorem 1.3, extends the linear theory of Lü and Wang [24] to nonlinear equations and resolve the higher-dimensional extension left open by Zhang et al. [33, Remark 1.4]. As a special case, we also ovtain the null controllability for a class of semilinear system with only one drift control; see Theorem 1.8. For superlinear nonlinearities, we further establish a local controllability result; see Theorem 1.14. To the best of our knowledge, this work seems to be the first to investigate almost-sure controllability for forward superlinear SPDEs; see [15] for a recent study on backward superlinear SPDEs based on a distinct approach.

1.1. **Comparison with existing results.**

1.1.1. *Carleman estimates and the linear stochastic theory.* Carleman estimates give adjoint observability and hence linear null controllability by duality [6, 16]. For deterministic fourth-order equations, Gao [7] treated negative-order source terms and nonlinearities involving spatial derivatives, while Guerrero and Kassab [10] developed multidimensional estimates and quantitative control bounds. For the clamped problem, Kassab [17] treated full Hessian dependence and growth beyond global Lipschitz continuity. Under Navier boundary conditions, Li and You [19] established controllability for nonlinearities that are globally Lipschitz in the state, gradient, and Hessian. In the stochastic setting, the control selection must also preserve adaptedness. For stochastic second-order equations, Tang and Zhang [30] derived a pointwise identity and an adjoint Carleman estimate to obtain controllability with a diffusion control. Liu [20] subsequently obtained estimates allowing spatially L^2 diffusion terms and H^{-1} drift sources, while Baroun et al. [3] obtained estimates for distributional sources that accommodate general convection terms. For one-dimensional stochastic fourth-order equations, see [8]. Lü and Wang [24] established multidimensional null controllability with an interior drift control

and a full-domain diffusion control. Their full-Hessian Carleman estimate is the starting point for our estimates under both boundary conditions.

1.1.2. *Semilinear equations and quantitative contraction arguments.* The closest semilinear result is due to Zhang et al. [33], who considered an equation on an interval with a drift that is globally Lipschitz in (y, y_x, y_{xx}). They left the higher-dimensional case open because of the difficulty of deriving the required Carleman estimates. Their contraction argument uses an H^{-2}-source estimate and weighted second-derivative bounds with spatially L^2 diffusion. In the present paper, the estimates cover tensor sources and the full Hessian under both boundary conditions, with full-domain diffusion compensation.

Hernández-Santamaría et al. [13] developed a quantitative contraction approach to null controllability for semilinear parabolic SPDEs. Zhang et al. extended this approach to gradient-dependent nonlinearities [32]. These works and [33] use Carleman weights that do not degenerate initially, thereby controlling the adjoint trace at $t = 0$ and obtaining contraction. Our argument combines a shifted-interval Carleman estimate with a weighted energy estimate at $t = 0$, whose parameter is chosen independently. Normalization on the overlap preserves initial-time nondegeneracy and allows the two parameters to be chosen sufficiently large for contraction.

1.1.3. *Null controllability with a drift control alone.* Null controllability by a drift control alone has been established under special structural conditions; to our knowledge, the general question remains open. For the stochastic heat equation with spatially constant scalar multiplicative noise, Lü [23] proved null controllability by combining spectral observability with time iteration; see also [31]. The abstract framework of [22] covers fourth-order equations with this type of noise, while [25] treats a second-order equation with spatially analytic noise on a torus. Scalar stochastic exponentials have also been used in control problems; see, for instance, [2, Theorem 5.1 and Remark 5.2]. Theorem 1.8 extends the scalar multiplication argument to diffusion operators of the form $G(\omega, t, z) = b(\omega, t, z)z$, where b is a bounded scalar coefficient and G is globally Lipschitz on $L^2(\mathcal{O})$. For deterministic, positively homogeneous full-Hessian drifts, the prescribed diffusion is recovered by combining a Lipschitz deterministic control selection with a positive scalar SDE amplitude. Conditional moment estimates then yield global null controllability for square-integrable random data. Example 1.10 uses saturated dependence on the squared $L^2(\mathcal{O})$-norm. State-independent scalar coefficients also give a Lipschitz stochastic control selection (Corollary 1.11). Adapted stochastic constructions and truncation methods provide another approach [12, 14]. In particular, [12, Theorem 1.2] establishes almost-sure null controllability by a drift control for an equation whose nonlinear coefficients are truncated using a pathwise trajectory norm. Theorem 1.3 of that work gives statistical local controllability for the original equation; in one space dimension, the diffusion may include a pointwise polynomial nonlinearity. Theorem 1.8 uses a current-state scalar coefficient multiplying the entire spatial profile, without a trajectory cutoff.

1.1.4. *Superlinear drifts and limits of global controllability.* Hernández-Santamaría et al. [15] studied backward parabolic SPDEs using solution regularity; the corresponding forward

problem remains unresolved in that work. Theorem 1.14 gives adapted almost-sure null controls for small data in $L^\infty(\Omega;V)$ and separate superlinear state and derivative terms, including Hessian growth up to $1+4/n$. The linear control cost follows from [18, 26], while full noise compensation supplies the pathwise regularity required for the nonlinear argument.

Following [17], Proposition 8.1 uses a localized moment to bound the existence time of energy-class solutions for a superlinear example, for every square-integrable diffusion control. Thus polynomial growth alone does not ensure global null controllability.

1.2. **Statement of main results.** Fix a complete filtered probability space $(\Omega,\mathcal{F},\mathbb{F},\mathbb{P})$ carrying a standard one-dimensional Brownian motion $W=\{W(t)\}_{t\geq 0}$. We assume that $\mathcal{F}_0$ is independent of W and that $\mathbb{F}=\{\mathcal{F}_t\}_{t\geq 0}$ is the usual augmentation of the filtration generated by $\mathcal{F}_0$ and W. Let $\mathcal{P}$ be the corresponding progressive σ-field on $\Omega\times[0,T]$.

Set $H=L^2(\mathcal{O};\mathbb{R})$ and $H_0=L^2(\mathcal{O}_0;\mathbb{R})$. For a separable Hilbert space X, we denote by $L^2_{\mathbb{F}}(0,T;X)$ the space of progressively measurable X-valued processes z for which

$$\|z\|^2_{L^2_{\mathbb{F}}(0,T;X)}:=\mathbb{E}\int_0^T\|z(t)\|_X^2\,\mathrm{d}t<\infty.$$

Similarly, $L^2_{\mathbb{F}}(\Omega;C([0,T];H))$ denotes the space of adapted processes with continuous H-valued paths such that

$$\|z\|^2_{L^2_{\mathbb{F}}(\Omega;C([0,T];H))}:=\mathbb{E}\sup_{t\in[0,T]}\|z(t)\|_H^2<\infty.$$

The form domains and the bilinear form are defined by

$$V_N=H^2(\mathcal{O})\cap H_0^1(\mathcal{O}),\qquad V_C=H_0^2(\mathcal{O}),\qquad a(y,z)=\int_{\mathcal{O}}\Delta y\,\Delta z\,\mathrm{d}x. \tag{1.2}$$

The Navier and clamped realizations A_N and A_C of Δ^2 are the nonnegative self-adjoint operators associated with a on V_N and V_C, respectively. Throughout the paper, we fix $A\in\{A_N,A_C\}$ and write $V=D(A^{1/2})$, equipped with the norm

$$\|y\|_V=\|A^{1/2}y\|_H=\|\Delta y\|_H.$$

The form domain and its associated operator encode the boundary conditions. We write $F[z]=F(\omega,t,\cdot,z,\nabla z,D^2z)$ and use the same convention for $g[z]$.

1.2.1. *Globally Lipschitz drifts.* For globally Lipschitz drifts, the diffusion is subject to the following energy coercivity condition.

Assumption 1.1. The maps $F,g:\Omega\times[0,T]\times\mathcal{O}\times\mathbb{R}\times\mathbb{R}^n\times\mathbb{R}^{n\times n}\to\mathbb{R}$ are $\mathcal{P}\otimes\mathcal{B}(\mathcal{O}\times\mathbb{R}\times\mathbb{R}^n\times\mathbb{R}^{n\times n})$-measurable.

(i) There exist deterministic constants $L_0,L_1,L_2\geq 0$ such that, outside a single $\mathbb{P}\otimes\mathrm{d}t\otimes\mathrm{d}x$-null set, $F(\omega,t,x,0,0,0)=0$ and, for all state arguments,

$$|F(r,p,X)-F(\widetilde{r},\widetilde{p},\widetilde{X})|\leq L_0|r-\widetilde{r}|+L_1|p-\widetilde{p}|+L_2|X-\widetilde{X}|. \tag{1.3}$$

(ii) The amplitude $g_* = g(\omega, t, x, 0, 0, 0)$ belongs to $L^2_{\mathbb{F}}(0,T;H)$. Outside a single $\mathbb{P}\otimes \mathrm{d}t$-null set, the map $g[z] = g(\omega, t, \cdot, z, \nabla z, D^2 z)$ is well defined from V to H and satisfies

$$\|g[z] - g[w]\|_H^2 \le a_g \|z-w\|_V^2 + b_g \|z-w\|_H^2, \qquad z, w \in V, \tag{1.4}$$

with deterministic $0 \le a_g < 2$ and $b_g \ge 0$.

Remark 1.2. In the present setting, the restriction $0 \le a_g < 2$ appears necessary for well-posedness of the controlled system. An example illustrating this restriction is given in Appendix A.

Theorem 1.3. *Let $A \in \{A_N, A_C\}$ and suppose that Assumption 1.1 holds. For every $T > 0$ and $y_0 \in L^2(\Omega, \mathcal{F}_0; H)$, there exists a control $(u, v) \in \mathcal{U} := L^2_{\mathbb{F}}(0,T;H_0) \times L^2_{\mathbb{F}}(0,T;H)$ such that the unique variational solution $y \in \mathcal{H} := L^2_{\mathbb{F}}(\Omega; C([0,T];H)) \cap L^2_{\mathbb{F}}(0,T;V)$ of (1.1) satisfies $y(T) = 0$, $\mathbb{P}$-a.s. in H. Moreover, if we set*

$$\Lambda := 1 + L_0 + L_1^{4/3} + L_2^2,$$

the controlled trajectory and the controls satisfy

$$\|y\|_{\mathcal{H}}^2 + \|(u,v)\|_{\mathcal{U}}^2 \le C(1+b_g) e^{C\Lambda}\, \mathbb{E}\|y_0\|_H^2 + 2\|g_*\|^2_{L^2_{\mathbb{F}}(0,T;H)}. \tag{1.5}$$

The state and controls can be selected by a globally Lipschitz map $\Gamma : L^2(\Omega, \mathcal{F}_0; H) \to \mathcal{H} \times \mathcal{U}$, with $\Gamma(y_0) = (y, u, v)$, $\Gamma(0) = (0, 0, -g_*)$, and

$$\|\Gamma(y_{0,1}) - \Gamma(y_{0,2})\|^2_{\mathcal{H}\times\mathcal{U}} \le C(1+b_g) e^{C\Lambda}\, \mathbb{E}\|y_{0,1} - y_{0,2}\|_H^2. \tag{1.6}$$

Here C depends only on $\mathcal{O}$, $\mathcal{O}_0$, T, and the boundary realization, and is uniform in a_g.

Remark 1.4. Since the diffusion control v acts throughout $\mathcal{O}$, we take the total diffusion $U := g[y] + v$ as a new control. We will first prove Theorem 1.3 for the system with diffusion term $U\,\mathrm{d}W_t$. For the resulting triple (y, u, U), Assumption 1.1(ii) ensures that the reconstructed control $v := U - g[y]$ belongs to $L^2_{\mathbb{F}}(0,T;H)$; this recovers the original equation. The same full-domain compensation was used in [33, Remark 1.2]; see also [13, 32].

Example 1.5 (A bounded nonlinear drift)**.** Let a_0, a_1, a_2 be bounded progressively measurable scalar fields, and let b, C be bounded progressively measurable vector and symmetric matrix fields. Then

$$F(\omega, t, x, r, p, X) = a_0 \tanh r + a_1 \tanh(b \cdot p) + a_2 \tanh(C : X)$$

satisfies Assumption 1.1(i). Measurability and vanishing at zero are immediate; $|\tanh s - \tanh t| \le |s - t|$ and Cauchy–Schwarz give $L_0 = \|a_0\|_\infty$, $L_1 = \|a_1 b\|_\infty$ and $L_2 = \|a_2 C\|_\infty$. If C is not a scalar multiple of the identity, the Hessian dependence in the last term cannot in general be expressed through Δy alone.

Remark 1.6 (A pointwise sufficient condition)**.** Assume that g_* satisfies the stated measurability and integrability conditions, and suppose that

$$|g(r,p,X) - g(\widetilde{r}, \widetilde{p}, \widetilde{X})| \le G_0|r - \widetilde{r}| + G_1|p - \widetilde{p}| + G_2|X - \widetilde{X}|.$$

Here the constants G_j are deterministic and nonnegative. Let $\|D^2 z\|_H \le C_{\mathcal{O},2}\|z\|_V$ on V. If $C_{\mathcal{O},2}^2 G_2^2 < 2$, then (1.4) holds. To verify this assertion, set $e = z - w$ and integrate the pointwise increment bound to obtain

$$\|g[z] - g[w]\|_H \le G_0\|e\|_H + G_1\|\nabla e\|_H + G_2\|D^2 e\|_H.$$

Since $e \in H_0^1(\mathcal{O}) \cap H^2(\mathcal{O})$ for either realization, integration by parts gives $\|\nabla e\|_H^2 = -(e, \Delta e)_H \le \|e\|_H\|e\|_V$. By Young's inequality, we first separate the Hessian term and then apply the interpolation bound to the gradient term. For any $\epsilon, \delta > 0$, we obtain

$$\|g[z] - g[w]\|_H^2 \le \big((1+\epsilon)C_{\mathcal{O},2}^2 G_2^2 + \delta\big)\|e\|_V^2 + C_{\epsilon,\delta}(G_0^2 + G_1^4)\|e\|_H^2, \quad e = z - w.$$

We conclude by choosing $\epsilon, \delta > 0$ so that the first coefficient is less than 2. In particular, G_0, G_1 need not be small. For $g[y] = g_* + \sigma\Delta y$, it suffices to assume $\|\sigma\|_\infty^2 < 2$, and no spatial derivatives of the coefficients are required.

1.2.2. *A single drift control.* We now turn to an H-valued diffusion operator, which need not be a pointwise function of the state and its spatial derivatives.

Assumption 1.7. (i) The deterministic drift F has the measurability specified in Assumption 1.1 and satisfies Assumption 1.1(i). It is positively homogeneous of degree one: for a.e. (t, x), every $(r, p, X) \in \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^{n\times n}$ and $c > 0$,

$$F(t, x, cr, cp, cX) = cF(t, x, r, p, X).$$

(ii) The diffusion coefficient has the form

$$G(\omega, t, z) = b(\omega, t, z)z,$$

where the map $b : \Omega \times [0, T] \times H \to \mathbb{R}$ is $\mathcal{P} \otimes \mathcal{B}(H)$-measurable. Assume that there exist deterministic constants $B_0, L_G \ge 0$ such that, outside a single $\mathbb{P} \otimes \mathrm{d}t$-null set, for every $z, \widetilde{z} \in H$,

$$|b(\omega, t, z)| \le B_0, \qquad \|G(\omega, t, z) - G(\omega, t, \widetilde{z})\|_H \le L_G\|z - \widetilde{z}\|_H.$$

Under this assumption, the controlled equation reads

$$\begin{cases} \mathrm{d}y + Ay\,\mathrm{d}t = \big(F[y] + Bu\big)\,\mathrm{d}t + G(t, y)\,\mathrm{d}W_t, \quad 0 < t < T, \\ y(0) = y_0, \end{cases} \tag{1.7}$$

where A specifies the boundary conditions and $B : H_0 \to H$ is the extension-by-zero operator. We then have the following result.

Theorem 1.8. *Let $A \in \{A_N, A_C\}$ and suppose that Assumption 1.7 holds. For every $T > 0$ and $y_0 \in L^2(\Omega, \mathcal{F}_0; H)$, there exists a control $u \in L^2_{\mathbb{F}}(0, T; H_0)$ such that the unique variational solution $y \in \mathcal{H}$ of (1.7) satisfies $y(T) = 0$ almost surely in H and*

$$\|y\|_{\mathcal{H}}^2 + \|u\|_{L^2_{\mathbb{F}}(0,T;H_0)}^2 \le Ce^{B_0^2 T + C\Lambda}\mathbb{E}\|y_0\|_H^2, \tag{1.8}$$

where $\Lambda = 1 + L_0 + L_1^{4/3} + L_2^2$, and C depends only on $\mathcal{O}, \mathcal{O}_0, T$ and the boundary realization.

Example 1.9 (Bellman–Isaacs and norm drifts)**.** Let a_{ij}, b_{ij}, C_{ij} be bounded deterministic measurable scalar, vector and matrix fields, respectively. The finite Isaacs operator

$$F(t,x,r,p,X) = \max_{1\le i\le m}\min_{1\le j\le k}\{a_{ij}(t,x)r + b_{ij}(t,x)\cdot p + C_{ij}(t,x) : X\}$$

satisfies the drift conditions in Assumption 1.7. One can choose the Lipschitz constants as $L_0 = \max_{i,j}\|a_{ij}\|_\infty$, $L_1 = \max_{i,j}\|b_{ij}\|_\infty$, and $L_2 = \max_{i,j}\|C_{ij}\|_\infty$, with the Frobenius matrix norm. The case $k = 1$ gives a Bellman maximum. Such structures also occur in second-order equations [27]; here the Hessian term is lower order relative to the biharmonic principal part.

One can also take $F(r,p,X) = \kappa|X|$ for any $\kappa \ge 0$. For $\kappa > 0$, this drift is nonlinear and is not differentiable at $X = 0$, while $|\kappa|X| - \kappa|\widetilde{X}|| \le \kappa|X - \widetilde{X}|$.

Example 1.10 (Saturated nonlinear diffusion)**.** For any $\sigma \in \mathbb{R}\setminus\{0\}$, consider

$$G_\sigma(z) = \sigma\frac{\|z\|_H^2}{1+\|z\|_H^2}\,z, \qquad z \in H.$$

The scalar coefficient is bounded by $|\sigma|$. With $s = \|z\|_H^2$, the Fréchet derivative of the diffusion operator is

$$DG_\sigma(z)h = \sigma\frac{s}{1+s}h + \frac{2\sigma(z,h)_H}{(1+s)^2}z.$$

The formula holds at $z = 0$ and gives $\|DG_\sigma(z)\|_{\mathcal{L}(H)} \le 3|\sigma|/2$. We have therefore verified Assumption 1.7(ii) with $B_0 = |\sigma|$ and $L_G = 3|\sigma|/2$. For example, the equation

$$\mathrm{d}y + \Delta^2 y\,\mathrm{d}t = \big(\kappa|D^2 y| + \mathbf{1}_{\mathcal{O}_0}u\big)\,\mathrm{d}t + \sigma\frac{\|y(t)\|_H^2}{1+\|y(t)\|_H^2}y\,\mathrm{d}W_t$$

with either boundary condition is globally null controllable by a drift control alone, for every $\kappa \ge 0$ and $\sigma \in \mathbb{R}$. The bound (1.8) becomes

$$\|y\|_{\mathcal{H}}^2 + \|u\|_{L^2_{\mathbb{F}}(0,T;H_0)}^2 \le Ce^{\sigma^2 T + C(1+\kappa^2)}\mathbb{E}\|y_0\|_H^2.$$

Corollary 1.11. *Assume that the drift satisfies Assumption 1.7(i). Let β_0 be a fixed progressively measurable real-valued process satisfying $|\beta_0| \le B_0$ almost everywhere, and set*

$$G(\omega,t,z) = \beta_0(\omega,t)z.$$

Then there exists a globally Lipschitz selection

$$\Gamma_{\mathrm{sc}} : L^2(\Omega,\mathcal{F}_0;H) \longrightarrow \mathcal{H}\times L^2_{\mathbb{F}}(0,T;H_0)$$

such that $\Gamma_{\mathrm{sc}}(0) = (0,0)$ and, for every initial state y_0, the pair $(y,u) = \Gamma_{\mathrm{sc}}(y_0)$ satisfies the conclusions of Theorem 1.8.

Remark 1.12. A related null-controllability result for the stochastic heat equation, with a single control acting on the drift, was proved by Lü [23]. Lü used spectral observability and time iteration. Here we select deterministic nonlinear controls by weighted contraction and transfer them to the prescribed diffusion through a scalar stochastic amplitude.

1.2.3. *Superlinear drifts.* For $j = 0, 1, 2$, define

$$p_j^* = \begin{cases} (n+4)/(n-4+2j), & n-4+2j > 0, \\ +\infty, & n-4+2j \le 0, \end{cases}$$

where $+\infty$ means that every finite exponent is allowed. The corresponding bounds are

n	p_0^*	p_1^*	p_2^*
2	$+\infty$	$+\infty$	3
3	$+\infty$	7	$7/3$
4	$+\infty$	4	2
$n \ge 5$	$(n+4)/(n-4)$	$(n+4)/(n-2)$	$1+4/n$.

Assumption 1.13. Let $J \subset \{0,1,2\}$ be nonempty, and assume that $1 < p_j \le p_j^*$ for every $j \in J$, with all exponents finite. Set $E_0 = \mathbb{R}$, $E_1 = \mathbb{R}^n$, and $E_2 = \mathbb{R}^{n\times n}_{\mathrm{sym}}$. The drift in (1.1) has the form

$$F(\omega, t, x, y, \nabla y, D^2 y) = \mathcal{N}(y) := \sum_{j\in J} f_j(\omega, t, x, D^j y),$$

where f_j is measurable with respect to $\mathcal{P} \otimes \mathcal{B}(\mathcal{O} \times E_j)$ and, outside one $\mathbb{P} \otimes \mathrm{d}t \otimes \mathrm{d}x$-null set, satisfies

$$f_j(\omega, t, x, 0) = 0, \qquad |f_j(\omega, t, x, \xi) - f_j(\omega, t, x, \eta)| \le K_j\big(|\xi|^{p_j-1} + |\eta|^{p_j-1}\big)|\xi - \eta| \tag{1.9}$$

for all $\xi, \eta \in E_j$, with deterministic $K_j < \infty$.

Theorem 1.14. *Let $A \in \{A_N, A_C\}$, and suppose that Assumptions 1.13 and 1.1(ii) hold. For every $T > 0$, there exists a constant $\delta_T > 0$ such that, for any $y_0 \in L^\infty(\Omega, \mathcal{F}_0; V)$ satisfying*

$$d := \|y_0\|_{L^\infty(\Omega;V)} \le \delta_T,$$

there exist progressively measurable controls

$$(u, v) \in L^\infty\big(\Omega; L^2(0,T;H_0)\big) \times L^2_{\mathbb{F}}(0,T;H),$$

such that the corresponding state satisfies $y \in L^\infty(\Omega; \mathfrak{X}_T)$ and $y(T) = 0$ almost surely, where $\mathfrak{X}_T := C([0,T];V) \cap L^2(0,T;D(A)) \cap H^1(0,T;H)$. Moreover, there is a constant $C_T > 0$ such that

$$\|y\|_{L^\infty(\Omega;\mathfrak{X}_T)} + \|u\|_{L^\infty(\Omega;L^2(0,T;H_0))} \le C_T d, \tag{1.10}$$

$$\mathbb{E}\int_0^T \|v(t)\|_H^2 \,\mathrm{d}t \le C_T(1+b_g)d^2 + 2\|g_*\|^2_{L^2_{\mathbb{F}}(0,T;H)}. \tag{1.11}$$

The constants δ_T and C_T depend only on the domain, the control region, T, the boundary realization, the exponents, and the constants K_j.

Example 1.15 (Polynomial derivative terms)**.** Let $n = 2$ and consider

$$F(\omega, t, x, r, p, X) = a_0 r^3 + a_1|p|^2 + a_2(C : X)^3,$$

where a_0, a_1, a_2 and the symmetric matrix field C are bounded and progressively measurable. Here $C : X$ denotes the Frobenius inner product. The cubic and quadratic increment bounds

verify Assumption 1.13. One can take $p_0 = 3$, $p_1 = 2$, $p_2 = 3$ and $K_0 = \frac{3}{2}\|a_0\|_\infty$, $K_1 = \|a_1\|_\infty$, $K_2 = \frac{3}{2}\|a_2\|_\infty \|C\|_\infty^3$. If the diffusion also satisfies Assumption 1.1(ii), Theorem 1.14 applies to sufficiently small initial data.

For the pure cubic reaction $F = a_0 r^3$, the same theorem applies in every dimension $2 \le n \le 8$, including the endpoint $n = 8$. Indeed, for $n > 4$, $3 \le (n+4)/(n-4)$ is equivalent to $n \le 8$; for $n \le 4$, every finite reaction exponent is admissible.

1.3. **Main difficulties and proof strategy.** The main difficulty lies in obtaining full-Hessian estimates strong enough to construct adapted nonlinear controls. Derivative terms produce distributional adjoint sources, spatially L^2 diffusion prevents direct H^2 Itô estimates, and spatial compactness alone does not yield compactness in the probability variable. We therefore use quantitative estimates, duality and contraction [13, 32, 33]. A weighted energy estimate controls the initial trace with a parameter independent of the one in the shifted-interval Carleman estimate (cf. [24]).

Weak sources and full-Hessian estimates. Under duality, the derivative-dependent terms become V'-sources of the form

$$\Phi_0 + \operatorname{div} \Phi_1 + \sum_{i,j} \partial_{ij} (\Phi_2)_{ij}.$$

The L^2-source estimate of [24] and an auxiliary null-control problem give reciprocal-weight Hessian bounds without differentiating the diffusion. The positive Hessian contribution from quadratic variation absorbs the negative gradient term even with spatially L^2 noise. Combining these estimates with the clamped tangential trace bound, we obtain Theorem 3.8.

Initial-time estimate and quantitative contraction. The Carleman estimate on $(T/4, T)$ is combined with a weighted energy estimate near $t = 0$. Normalizing by the minimum of the Carleman state weight on the overlap controls the cutoff error; the normalization factor cancels in the weight ratios. For independent parameters s and κ, these ratios satisfy

$$\frac{q_{j,s,\kappa}}{b_{s,\kappa}} \le c_j s^{-6+2j} + \kappa^{-2+j/2}, \qquad j = 0, 1, 2.$$

For differences with equal initial states, the initial-data cost vanishes. Weighted duality selects adapted controls and identifies the state, its derivatives and its terminal value. The independent parameter choice in (5.4) gives contraction; (4.4) then yields the cost exponent $1 + L_0 + L_1^{4/3} + L_2^2$. The selected total diffusion U recovers the original control through $v = U - g[y]$.

Retaining nonlinear multiplicative noise. For deterministic, positively homogeneous drifts, we select the state and control through a Lipschitz map $y_0 \mapsto (w, h)$. For random initial data, this pair is $\mathcal{F}_0$-measurable. We then solve

$$\mathrm{d}M_t = M_t b(\omega, t, M_t w(t))\, \mathrm{d}W_t, \qquad M_0 = 1.$$

The scalar equation is well posed by global Lipschitz continuity of $G(z) = b(z)z$, even if b is not Lipschitz in z. Boundedness of b ensures positivity of M and conditional second-moment bounds. Homogeneity gives $y = Mw$, $u = Mh$ in (1.7). Since M is spatially constant, it preserves the boundary conditions. The conditional moment estimates bound the cost

without requiring the amplitude to be independent of the initial datum. For $b = \beta_0(\omega, t)$, the same amplitude can be used for different initial states, which gives Corollary 1.11.

Superlinear growth and full noise compensation. A deterministic source operator supplies the pathwise H^4 regularity needed for polynomial Hessian terms. The source accumulated on each time interval is controlled on the next interval, so the construction preserves adaptedness. Terminal weights dominate the short-time cost and satisfy $\rho_y^{p_j} \le \rho_f$; space–time interpolation gives contraction on a small $L^\infty(\Omega)$ ball. The stochastic equation is recovered by setting $v = -g[y]$.

A superlinear obstruction. A test function supported outside $\mathcal{O}_0$ removes the drift control, while square integrability removes the stochastic integral in expectation. For $F(y) = -|y|^{1+1/n}$, a sufficiently negative initial moment yields a superlinear inequality, which rules out null controllability and gives the existence-time bound in Section 8.

1.4. **Organization of the paper.** In Section 2, we establish well-posedness and duality. The Carleman estimates will be derived in Section 3, and linear null controls will be constructed in Section 4. Sections 5 and 6 contain the proofs of the global results. Section 7 is devoted to superlinear local null controllability, while Section 8 presents the counterexample.

2. Variational solutions and forward–backward duality

2.1. **Coefficient maps and variational spaces.** We define $A : V \to V'$ by (1.2), with $\langle Ay, z\rangle = a(y, z)$. Its realization in H is positive and self-adjoint, and elliptic regularity gives

$$\begin{aligned} &D(A_N) = \{y \in H^4(\mathcal{O}) : y|_{\partial\mathcal{O}} = \Delta y|_{\partial\mathcal{O}} = 0\}, && A_N y = \Delta^2 y, \\ &D(A_C) = H^4(\mathcal{O}) \cap H_0^2(\mathcal{O}), && A_C y = \Delta^2 y. \end{aligned} \tag{2.1}$$

In the Gelfand triple $V \hookrightarrow H \hookrightarrow V'$, the form domain V encodes the boundary conditions; H data and elements of V_N need not admit a classical trace of Δy.

Lemma 2.1. *Let $E \subset \mathcal{O}$ be measurable, set $H_E = L^2(E)$, and let $q(\omega, t, x, r, p, X)$ be measurable with respect to*

$$\mathcal{P} \otimes \mathcal{B}(E) \otimes \mathcal{B}(\mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^{n\times n}).$$

For $z \in V$, let $q[z]$ denote q evaluated at $(z, \nabla z, D^2 z)|_E$. Assume $q[0] \in L^2_{\mathbb{F}}(0, T; H_E)$ and that there is a single $\mathbb{P} \otimes \mathrm{d}t$-null set N such that, for every $(\omega, t) \notin N$, the map $z \mapsto q[z](\omega, t)$ is well defined from V to H_E and satisfies

$$\|q[z] - q[w]\|^2_{H_E} \le a\|z - w\|^2_V + b\|z - w\|^2_H \qquad \text{for all } z, w \in V,$$

where $a, b \ge 0$ are deterministic constants.

Then $q[\cdot]$ has an H_E-valued, $\mathcal{P} \otimes \mathcal{B}(V)$-measurable version agreeing with $q[z]$ for every $z \in V$ outside a common base null set. For any $Y, Z \in L^2_{\mathbb{F}}(0, T; V)$, $q[Y]$ and $q[Z]$ belong to $L^2_{\mathbb{F}}(0, T; H_E)$ and satisfy

$$\|q[Y] - q[Z]\|^2_{L^2_{\mathbb{F}}(0,T;H_E)} \le a\|Y - Z\|^2_{L^2_{\mathbb{F}}(0,T;V)} + b\|Y - Z\|^2_{L^2_{\mathbb{F}}(0,T;H)}. \tag{2.2}$$

Proof. For fixed $z \in V$, choose Borel representatives of z, ∇z and $D^2 z$ on E. Then $q[z]$ is $\mathcal{P} \otimes \mathcal{B}(E)$-measurable. By simple-function approximation and separability of H_E, we obtain a progressive H_E-valued version, extended by zero where its spatial norm is infinite. The increment bound with $w = 0$ and the assumption $q[0] \in L^2_{\mathbb{F}}(0,T;H_E)$ show that this extension changes the process only on a base null set.

Choose a dense sequence (z_ℓ) in V and, for each m, let $\pi_m(z)$ be the first z_ℓ satisfying $\|z_\ell - z\|_V < 1/m$. The map $\pi_m : V \to \{z_\ell\}$ is Borel. Progressive H_E-valued versions of $q[z_\ell]$ make $q[\pi_m z]$ $\mathcal{P} \otimes \mathcal{B}(V)$-measurable. Outside the original exceptional set and the countable union of version null sets, the increment bound and $V \hookrightarrow H$ give continuity of $z \mapsto q[z]$ from V to H_E. It follows that $q[\pi_m z] \to q[z]$ for every $z \in V$ on that set. Completeness and separability of H_E yield a jointly measurable limit, set to zero where convergence fails; it agrees with every $q[z]$ off one base null set.

Composing with progressive V-valued representatives of Y and Z gives progressive H_E-valued processes; integration then yields (2.2). Moreover,

$$\|q[Y]\|^2_{H_E} \le 2\|q[0]\|^2_{H_E} + 2a\|Y\|^2_V + 2b\|Y\|^2_H,$$

and the same bound proves square integrability of $q[Z]$. Both compositions are independent of the chosen process and weak-derivative representatives. □

2.2. **Well-posedness of the controlled system.**

Proposition 2.2. *Suppose Assumption 1.1 holds, and write $F[y] = F(y, \nabla y, D^2 y)$. For every $y_0 \in L^2(\Omega, \mathcal{F}_0; H)$ and $(u, v) \in \mathcal{U}$, the system*

$$\begin{cases} \mathrm{d}y + Ay\,\mathrm{d}t = (F[y] + Bu)\,\mathrm{d}t + (g[y] + v)\,\mathrm{d}W_t, \\ y(0) = y_0 \end{cases} \tag{2.3}$$

has a unique variational solution $y \in \mathcal{H}$. In particular, for every $\psi \in V$ and $t \in [0,T]$,

$$\begin{aligned}(y(t),\psi)_H = (y_0,\psi)_H - \int_0^t a(y(\tau),\psi)\,\mathrm{d}\tau + \int_0^t (F[y](\tau) + Bu(\tau),\psi)_H\,\mathrm{d}\tau \\ + \int_0^t (g[y](\tau) + v(\tau),\psi)_H\,\mathrm{d}W_\tau, \quad \mathbb{P}\text{-a.s.}\end{aligned}$$

With coefficients fixed, write the solution, initial-data and control differences as δy, δy_0, δu, and δv, respectively. They satisfy

$$\|\delta y\|^2_{\mathcal{H}} \le C_E \left(\mathbb{E}\|\delta y_0\|^2_H + \|\delta u\|^2_{L^2_{\mathbb{F}}(0,T;H_0)} + \|\delta v\|^2_{L^2_{\mathbb{F}}(0,T;H)} \right). \tag{2.4}$$

Here C_E depends only on $\mathcal{O}$, T, the boundary realization, L_0, L_1, L_2, a_g, and b_g, and is independent of g_*; it may diverge as $a_g \uparrow 2$. Increasing C_E if needed, we have

$$\|y\|^2_{\mathcal{H}} \le C_E \left(\mathbb{E}\|y_0\|^2_H + \|u\|^2_{L^2_{\mathbb{F}}(0,T;H_0)} + \|v\|^2_{L^2_{\mathbb{F}}(0,T;H)} + \|g_*\|^2_{L^2_{\mathbb{F}}(0,T;H)} \right). \tag{2.5}$$

Proof. The proof is given in Appendix B. □

2.3. **Backward equation and duality.**

Lemma 2.3. *Let $z_T \in L^2(\Omega, \mathcal{F}_T; H)$ and $\Phi \in L^2_{\mathbb{F}}(0,T;H)$. Then the backward equation*

$$\mathrm{d}z - Az\,\mathrm{d}t = \Phi\,\mathrm{d}t + Z\,\mathrm{d}W_t, \qquad z(T) = z_T, \tag{2.6}$$

admits a unique variational solution $(z, Z) \in \mathcal{H} \times L^2_{\mathbb{F}}(0,T;H)$. For every $\psi \in V$, the identity holds almost surely for all $t \in [0,T]$:

$$(z(t),\psi)_H = (z_T,\psi)_H - \int_t^T \big[a(z(\tau),\psi) + (\Phi(\tau),\psi)_H\big]\,\mathrm{d}\tau - \int_t^T (Z(\tau),\psi)_H\,\mathrm{d}W_\tau.$$

Moreover,

$$\|z\|^2_{\mathcal{H}} + \|Z\|^2_{L^2_{\mathbb{F}}(0,T;H)} \le C\left(\mathbb{E}\|z_T\|^2_H + \|\Phi\|^2_{L^2_{\mathbb{F}}(0,T;H)}\right), \tag{2.7}$$

where C depends only on $\mathcal{O}$, T, and the boundary realization.

Let $y_0 \in L^2(\Omega, \mathcal{F}_0; H)$, $\phi, U \in L^2_{\mathbb{F}}(0,T;H)$, and $u \in L^2_{\mathbb{F}}(0,T;H_0)$. If Y solves

$$\mathrm{d}Y + AY\,\mathrm{d}t = (\phi + Bu)\,\mathrm{d}t + U\,\mathrm{d}W_t, \qquad Y(0) = y_0, \tag{2.8}$$

then

$$\begin{aligned}\mathbb{E}(Y(T), z_T)_H = \mathbb{E}(y_0, z(0))_H &+ \mathbb{E}\int_0^T \big[(\phi, z)_H + (Y,\Phi)_H + (U,Z)_H\big]\,\mathrm{d}t \\ &+ \mathbb{E}\int_0^T (u, z|_{\mathcal{O}_0})_{H_0}\,\mathrm{d}t.\end{aligned} \tag{2.9}$$

Proof. Step 1: Spectral approximation. Let (e_k) be an orthonormal eigenbasis of A, $Ae_k = \mu_k e_k$, and let P_N denote the projection onto the first N modes. Set $\Phi_k = (\Phi, e_k)_H$. The scalar equation

$$\mathrm{d}z_k = (\mu_k z_k + \Phi_k)\,\mathrm{d}t + Z_k\,\mathrm{d}W_t, \qquad z_k(T) = (z_T, e_k)_H,$$

admits the representation

$$z_k(t) = e^{\mu_k t}\mathbb{E}\left[e^{-\mu_k T}(z_T, e_k)_H - \int_t^T e^{-\mu_k \tau}\Phi_k(\tau)\,\mathrm{d}\tau \,\middle|\, \mathcal{F}_t\right].$$

The integrand $e^{-\mu_k t}Z_k(t)$ is determined by martingale representation.

In the augmented filtration of W and the independent initial σ-field, every square-integrable martingale has an $\mathcal{F}_0$-measurable initial value and an integral representation against W, with no orthogonal part. Independence and Brownian representation first give this for products of bounded initial and Brownian cylinder variables. Density and Itô's isometry then give the general case. The first N modes define (z_N, Z_N), where $z_N \in S^2(H)$ and $Z_N \in L^2_{\mathbb{F}}(0,T;P_NH)$.

Step 2: Estimates and convergence. We suppress the projection indices. Itô's formula from t to T gives

$$\mathbb{E}\|z(t)\|^2_H + \mathbb{E}\int_t^T \big(2\|z\|^2_V + \|Z\|^2_H\big)\,\mathrm{d}\tau = \mathbb{E}\|z_T\|^2_H - 2\mathbb{E}\int_t^T (\Phi, z)_H\,\mathrm{d}\tau.$$

The continuous embedding $V \hookrightarrow H$ gives

$$2|(\Phi, z)_H| \le \|z\|^2_V + C_{\mathcal{O}}\|\Phi\|^2_H.$$

These bounds control the mean energy and the $L^2_{\mathbb{F}}(0,T;V)\times L^2_{\mathbb{F}}(0,T;H)$ norm.

For the maximal estimate, set

$$M_t=\int_0^t (z(\tau),Z(\tau))_H\,\mathrm{d}W_\tau.$$

Since $\sup_{t\le T}|M_T-M_t|\le 2\sup_{t\le T}|M_t|$, the BDG and Young inequalities give

$$2\mathbb{E}\sup_{t\le T}|M_T-M_t|\le \frac{1}{4}\|z\|^2_{S^2(H)}+C\|Z\|^2_{L^2_{\mathbb{F}}(0,T;H)}.$$

Combining this with the mean-energy bound, we absorb the maximal term and obtain (2.7), uniformly in N.

The difference estimate for data $(P_N-P_M)z_T$ and $(P_N-P_M)\Phi$ gives strong convergence of (z_N,Z_N) in $\mathcal{H}\times L^2_{\mathbb{F}}(0,T;H)$. Pass to the limit as in Step 1 of the proof of Proposition 2.2. Uniqueness follows from the same difference estimate.

Step 3: Duality. On the same spectral subspace, Itô's product formula gives

$$\begin{aligned}\mathrm{d}(Y_N,z_N)_H=&\big[(P_N(\phi+Bu),z_N)_H+(Y_N,P_N\Phi)_H+(P_NU,Z_N)_H\big]\,\mathrm{d}t\\&+\big[(Y_N,Z_N)_H+(z_N,P_NU)_H\big]\,\mathrm{d}W_t.\end{aligned}$$

Symmetry of a cancels the form terms; covariation contributes $(P_NU,Z_N)_H$ with a positive sign. By BDG, we bound the expected martingale suprema by

$$C\big(\|Y_N\|_{S^2(H)}\|Z_N\|_{L^2_{\mathbb{F}}(0,T;H)}+\|z_N\|_{S^2(H)}\|P_NU\|_{L^2_{\mathbb{F}}(0,T;H)}\big).$$

The martingales have zero expectation. By strong energy convergence and Cauchy–Schwarz, we pass to the limit in the pairings, and $(Bu,z)_H=(u,z|_{\mathcal{O}_0})_{H_0}$ gives (2.9). □

The same results hold on any deterministic interval $[a,T]$, using

$$W^a(t)=W(a+t)-W(a),\qquad \mathcal{F}^a_t=\mathcal{F}_{a+t},\qquad 0\le t\le T-a,$$

with initial σ-field $\mathcal{F}_a$.

3. A Carleman estimate with distributional sources

We will extend the H-valued source estimate to distributional sources through auxiliary null controls and weighted derivative bounds. Section 4 supplies the initial-time energy estimate.

3.1. Weights and the Carleman estimate with H-valued sources.

Set

$$a=\frac{T}{4},\qquad b=\frac{T}{2},\qquad c=\frac{3T}{4},\qquad J_t=[b,c],$$

and

$$I_*=(a,T),\qquad Q_*=I_*\times\mathcal{O},\qquad Q_{0,*}=I_*\times\mathcal{O}_0,\qquad \Sigma_*=I_*\times\partial\mathcal{O}.$$

Choose a nonempty open set $\mathcal{O}_\sharp\Subset\mathcal{O}_0$ and an escape function $\eta\in C^\infty(\overline{\mathcal{O}})$ such that

$$\eta>0\text{ in }\mathcal{O},\qquad \eta=0\text{ on }\partial\mathcal{O},\qquad |\nabla\eta|>0\text{ on }\overline{\mathcal{O}\setminus\mathcal{O}_\sharp}.$$

Such a function exists by [6, Lemma 1.1]; see also [24, Eq. (1.4)]. Smoothness of the escape function permits differentiating the weight coefficients without further regularity of the data.

Let $M = \|\eta\|_{C(\overline{\mathcal{O}})}$ and fix $\lambda \geq \lambda_0$. For $(t,x) \in Q_*$, define

$$\xi_*(t,x) = \frac{e^{\lambda(2M+\eta(x))}}{\sqrt{(t-a)(T-t)}}, \qquad \alpha_*(t,x) = \frac{e^{\lambda(2M+\eta(x))} - e^{4\lambda M}}{\sqrt{(t-a)(T-t)}}, \qquad \theta_*(t,x) = e^{s\alpha_*(t,x)}. \tag{3.1}$$

Take $s \geq s_0 \geq 1$ and put $\sigma = s\lambda\xi_*$. Weights containing θ_*^2 are extended to $[0,T]$ by zero on $[0,a]$. Set

$$F_s = \theta_*^2, \quad B_s^* = s^6\lambda^8\xi_*^6\theta_*^2, \quad O_s^* = s^7\lambda^8\xi_*^7\theta_*^2, \quad R_s^* = e^{C_*\lambda M}s^4\lambda^4\xi_*^4\theta_*^2. \tag{3.2}$$

Choose C_* sufficiently large and keep λ fixed as s increases. For fixed s and λ, these weights are bounded on $[a,T] \times \overline{\mathcal{O}}$, positive on $(a,T) \times \overline{\mathcal{O}}$ and zero at $t = a$ and $t = T$; reciprocals are used only on I_*.

Remark 3.1 (Correction to the weighted identity)**.** In the identity of Lü and Wang [24], I and I_3 are defined in [24, Eqs. (2.3) and (2.7)], Ψ and Φ in [24, Eqs. (2.4) and (2.6)], and the fluxes V_1, V_2 and coefficients Θ_1, Θ_2 in [24, Theorem 2.2, pp. 1568–1569]. In this calculation, $\mathcal{I}_3$ denotes the $\mathrm{d}t$ density of I_3.

At a spectral cutoff for either realization, set $\ell = s\alpha_*$ and $v_g = \theta_* g$. The product rule gives

$$\partial_{ij} v_g = \theta_*\big[g_{ij} + \ell_i g_j + \ell_j g_i + (\ell_{ij} + \ell_i\ell_j)g\big].$$

The printed term $2\ell_j g_i$ in [24, Eq. (2.8)] differs from this formula for $n > 1$. Set

$$\mathbf{a} = \nabla\ell, \qquad \mathbf{b} = \nabla g, \qquad S = \theta_*^{-1} D^2 v_g, \qquad K = \mathbf{b}\otimes\mathbf{a} - \mathbf{a}\otimes\mathbf{b}.$$

The printed square uses $S + K$. Symmetry of S and skew-symmetry of K give

$$|S+K|^2 - |S|^2 = |K|^2 = 2\big(|\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a}\cdot\mathbf{b})^2\big) \geq 0.$$

We use the symmetric Hessian square and retain all deterministic terms, time differentials, fluxes V_1, V_2 and stochastic divergences.

The identities $I_{16}, I_{26}, I_{36}, I_{66}$ in [24, Appendix A, pp. 1586–1587] give $|\mathrm{d}(D^2 w)|^2$. Since $\mathrm{d}w$ has diffusion $-v_g$, its $\mathrm{d}t$ density is $|D^2 v_g|^2$. The corrected integral is

$$\mathscr{Q}_t(g) = \int_{\mathcal{O}} \big[|D^2 v_g|^2 - 2|\nabla\ell|^2|\nabla v_g|^2 - 4|\nabla\ell\cdot\nabla v_g|^2 + |\nabla\ell|^4|v_g|^2\big]\,\mathrm{d}x. \tag{3.3}$$

Set $K_s(t) = \|\nabla\ell(t,\cdot)\|^2_{L^\infty(\mathcal{O})}$ and write $\|\cdot\|_2 = \|\cdot\|_{L^2(\mathcal{O})}$. At either spectral cutoff, $v_g \in H^2(\mathcal{O}) \cap H_0^1(\mathcal{O})$. Integration by parts and $\|\Delta v_g\|_2 \leq \sqrt{n}\,\|D^2 v_g\|_2$ yield

$$\|\nabla v_g\|_2^2 = -(v_g, \Delta v_g)_H \leq \sqrt{n}\,\|v_g\|_2\|D^2 v_g\|_2.$$

By (3.3), we may discard the nonnegative zeroth-order term. Young's inequality then yields

$$\mathscr{Q}_t(g) \geq \|D^2 v_g\|_2^2 - 6K_s\|\nabla v_g\|_2^2 \geq \frac{1}{2}\|D^2 v_g\|_2^2 - 18nK_s^2\|v_g\|_2^2. \tag{3.4}$$

Now $\nabla\ell = \sigma\nabla\eta$ and $\sup_x \sigma(t,x)/\inf_x \sigma(t,x) \leq e^{\lambda M}$. Thus, for $(t,x) \in Q_*$,

$$K_s(t)^2 \leq \|\nabla\eta\|^4_{L^\infty(\mathcal{O})} e^{4\lambda M}\sigma(t,x)^4.$$

Since $v_g = \theta_* g$ and $R_s^* = e^{C_*\lambda M} F_s \sigma^4$, choosing $C_* \geq 4$ gives

$$\mathbb{E}\int_a^T \mathscr{Q}_t(g)\,\mathrm{d}t \geq -Ce^{4\lambda M}\mathbb{E}\int_{Q_*} F_s\sigma^4|g|^2\,\mathrm{d}x\,\mathrm{d}t \geq -C\mathbb{E}\int_{Q_*} R_s^*|g|^2\,\mathrm{d}x\,\mathrm{d}t.$$

The bound is uniform in s and the cutoff, without derivatives of g; hence we can remove the cutoff for either realization.

Proposition 3.2. *Let $A \in \{A_N, A_C\}$ and fix sufficiently large λ. There exist $s_0, C_0 > 0$ such that, for every $s \geq s_0$, every $z_T \in L^2(\Omega, \mathcal{F}_T; H)$ and every $\Phi \in L^2_{\mathbb{F}}(a,T;H)$, the variational solution (z,Z) of*

$$\begin{cases} \mathrm{d}z - Az\,\mathrm{d}t = \Phi\,\mathrm{d}t + Z\,\mathrm{d}W_t, \\ \qquad z(T) = z_T. \end{cases} \tag{3.5}$$

satisfies

$$\begin{aligned} \mathcal{B}_s^*(z) &:= \mathbb{E}\int_{Q_*} F_s\big(s^6\lambda^8\xi_*^6|z|^2 + s^4\lambda^6\xi_*^4|\nabla z|^2\big)\,\mathrm{d}x\,\mathrm{d}t \\ &\quad + \mathbb{E}\int_{Q_*} F_s\big(s^3\lambda^4\xi_*^3|\Delta z|^2 + s^2\lambda^4\xi_*^2|D^2 z|^2\big)\,\mathrm{d}x\,\mathrm{d}t \\ &\leq C_0\bigg(\mathbb{E}\int_{Q_{0,*}} O_s^*|z|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} F_s|\Phi|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} R_s^*|Z|^2\,\mathrm{d}x\,\mathrm{d}t\bigg). \end{aligned} \tag{3.6}$$

The constants depend only on $\mathcal{O}, \mathcal{O}_0, T$, the fixed weight and boundary realization, and not on s, z_T, Φ. Here $I_* = (a,T)$.

Proof. For Navier conditions, we apply [24, Theorem 1.8 and Section 3] to [24, Eq. (3.1)], with $h = z$, $f = -\Phi$, $g = -Z$ and (3.3)–(3.4). Time translation gives (3.1) on (a,T) with the filtration of Subsection 2.3. Proposition 3.3 gives the clamped estimate. The estimate extends to variational solutions by the spectral limit below, with bounds uniform in the cutoff and $s \geq s_0$ for fixed λ. □

3.2. **Clamped boundary conditions.** For smooth z, set $w = \theta_* z$. On $\partial\mathcal{O}$, the product rule gives

$$\begin{aligned} z = \Delta z = 0 \quad &\Longrightarrow \quad \Delta w = 2s\lambda\xi_*\partial_\nu\eta\,\partial_\nu w, \\ z = \partial_\nu z = 0 \quad &\Longrightarrow \quad w = \partial_\nu w = 0, \qquad \Delta w = \theta_*\Delta z. \end{aligned}$$

We therefore treat the Navier and clamped boundary terms separately. To control the clamped H^4 remainder in [24, Theorem 2.2], weighted by θ_b^2 with $\theta_b = \theta_*|_{\partial\mathcal{O}}$, we first estimate solutions with zero terminal data.

For either realization, let (h,G) solve variationally

$$\mathrm{d}h = Ah\,\mathrm{d}t + \Phi\,\mathrm{d}t + G\,\mathrm{d}W_t \quad \text{on } (a,T), \qquad h(T) = 0,$$

with $\Phi \in L^2_{\mathbb{F}}(a,T;H)$. The energy estimate gives h in the energy space and G in $L^2_{\mathbb{F}}(a,T;H)$. For the spectral projection P_N, set $h_N = P_N h$, $G_N = P_N G$ and $\Phi_N = P_N\Phi$. Itô's formula

applied to $\|A^{1/2}h_N\|_H^2$ on $[a,T]$, using $h_N(T)=0$, localization and Young's inequality, gives

$$\mathbb{E}\|A^{1/2}h_N(a)\|_H^2+\mathbb{E}\int_a^T\left(\|Ah_N\|_H^2+\|A^{1/2}G_N\|_H^2\right)\mathrm{d}t\le\mathbb{E}\int_a^T\|\Phi_N\|_H^2\,\mathrm{d}t. \tag{3.7}$$

Monotone convergence and $\Phi_N\to\Phi$ in $L^2_{\mathbb{F}}(a,T;H)$ give

$$h\in L^2_{\mathbb{F}}(a,T;D(A)),\qquad G\in L^2_{\mathbb{F}}(a,T;V),$$

with the same estimate. Energy convergence identifies the limits. Elliptic regularity and strict positivity of A imply

$$\|h\|_{H^4(\mathcal{O})}\le C\|Ah\|_H.$$

Multiplication by the time-dependent θ_b commutes with A, so the estimate applies to $\theta_b h$ whenever its transformed source lies in $L^2_{\mathbb{F}}(a,T;H)$.

For L^2 data in the shifted filtration of Subsection 2.3, $(P_Nz,P_N\Phi,P_NZ)$ converge in energy spaces by (2.7), with

$$P_Nz\longrightarrow z\quad\text{in } L^2_{\mathbb{F}}(a,T;H^2(\mathcal{O})).$$

At a fixed cutoff, the H norm controls all derivatives, so each martingale integrand K_t satisfies

$$\mathbb{E}\left(\int_a^T|K_t|^2\,\mathrm{d}t\right)^{1/2}\le C_{N,s,\lambda}\left(\mathbb{E}\sup_{t\in[a,T]}\|P_Nz(t)\|_H^2\right)^{1/2}\left(\mathbb{E}\int_a^T\|P_NZ(t)\|_H^2\,\mathrm{d}t\right)^{1/2}<\infty.$$

BDG removes localization in expected supremum norm, giving zero expectation with only square-integrable data. After obtaining bounds uniform in N, fix s,λ and pass to the limit using bounded weights and strong energy convergence. The constants used to justify these identities at a fixed cutoff may depend on N.

Proposition 3.3. *Let $A=A_C$ and fix a sufficiently large λ. There exist constants $s_0,C>0$, depending only on $\mathcal{O}$, $\mathcal{O}_0$, $T-a$, and the fixed spatial weight, such that, for every $z_T\in L^2(\Omega,\mathcal{F}_T;H)$ and $\Phi\in L^2_{\mathbb{F}}(a,T;H)$, the solution of (3.5) satisfies (3.6) for all $s\ge s_0$. Here C is independent of s, z_T, and Φ; for Z, only $Z\in L^2_{\mathbb{F}}(a,T;H)$ is required.*

Proof. At a spectral cutoff, we apply the corrected identity in [24, Theorem 2.2] and suppress the projection indices. All ensuing estimates are uniform in the cutoff.

Step 1: Boundary flux. Let h solve $-\mathrm{d}h+Ah\,\mathrm{d}t=f\,\mathrm{d}t+g\,\mathrm{d}W_t$ with terminal value $h_T\in L^2(\Omega,\mathcal{F}_T;H)$ and $f,g\in L^2_{\mathbb{F}}(a,T;H)$, and set $w=\theta_*h$. For (3.5), $h=z$, $f=-\Phi$, $g=-Z$. On $\partial\mathcal{O}$, set

$$k_b=s\lambda\xi_*|_{\partial\mathcal{O}},\quad \gamma=\partial_\nu\eta<0,\quad q_b=\Delta w,\quad r_b=\partial_\nu\Delta w.$$

Here k_b and θ_b depend only on time, and compactness of $\partial\mathcal{O}$ gives $-\gamma\ge\gamma_0>0$. Using $w=\nabla w=0$, we obtain $D^2w=q_b\nu\otimes\nu$ by tangential differentiation; the product rule then gives $q_b=\theta_b\Delta h$. In the weighted identity, $\Phi_1=-4k_b\gamma\nu$, $\Psi_2=2k_b^2\gamma^2$, $\Psi_3=4k_b^2\gamma^2\nu\otimes\nu$ and $\Phi_4=-4k_b^3\gamma^3\nu$ on the boundary. Thus the flux V_1 satisfies

$$V_1\cdot\nu=-2k_b\gamma|r_b|^2+2k_b\gamma|\nabla_{\partial\mathcal{O}}q_b|^2-10k_b^3\gamma^3|q_b|^2.$$

The $\nabla\Delta w$ terms give the first two contributions, while Hessian contractions give $(-4-16+8+4-2)k_b^3\gamma^3|q_b|^2$. The sign $\gamma<0$ makes the coefficients of $|r_b|^2$ and $|q_b|^2$ positive.

In V_2, only $\Theta_1 D^3wD^2w$ and $\Theta_2 D^2wD^2w$ survive, with $|\Theta_1|+|\Theta_2|\le C_\lambda k_b$. Tangentially differentiating $D^2w=q_b\nu\otimes\nu$ and using r_b for the normal derivative, including curvature terms, gives $|D^3w|\le C_{\mathcal{O}}(|r_b|+|\nabla_{\partial\mathcal{O}}q_b|+|q_b|)$ and

$$|V_2\cdot\nu|\le \delta k_b|r_b|^2+C_{\delta,\lambda}k_b(|\nabla_{\partial\mathcal{O}}q_b|^2+|q_b|^2).$$

Choose δ sufficiently small to absorb the $|r_b|^2$ term into V_1. Since $k_b\ge c_\lambda s$, absorb $k_b|q_b|^2$ into $k_b^3|q_b|^2$ by increasing s_0. Boundary interpolation and Young's inequality give, for every $\epsilon>0$,

$$k_b\|q_b\|^2_{H^1(\partial\mathcal{O})}\le \epsilon k_b^3\|q_b\|^2_{L^2(\partial\mathcal{O})}+C_\epsilon\|q_b\|^2_{H^{3/2}(\partial\mathcal{O})}\le \epsilon k_b^3\|q_b\|^2_{L^2(\partial\mathcal{O})}+C_\epsilon\theta_b^2\|h\|^2_{H^4(\mathcal{O})}.$$

The trace map $H^2(\mathcal{O})\to H^{3/2}(\partial\mathcal{O})$ applied to Δh gives the second inequality. For sufficiently small ϵ, there are $c_1,c_2>0$ such that

$$\int_{\partial\mathcal{O}}(V_1+V_2)\cdot\nu\,\mathrm{d}S\ge c_1k_b\|r_b\|^2_{L^2(\partial\mathcal{O})}+c_2k_b^3\|q_b\|^2_{L^2(\partial\mathcal{O})}-C_\lambda\theta_b^2\|h\|^2_{H^4(\mathcal{O})}.$$

We have obtained this bound uniformly in s and the spectral cutoff.

Set $\alpha_b=\alpha_*|_{\partial\mathcal{O}}$ and $\mathcal{R}_4(h)=\mathbb{E}\int_a^T\theta_b^2\|h\|^2_{H^4(\mathcal{O})}\,\mathrm{d}t$. For $v=\theta_b h$, we have

$$\mathrm{d}v-Av\,\mathrm{d}t=\big[-\theta_b f+s(\alpha_b)_t v\big]\,\mathrm{d}t-\theta_b g\,\mathrm{d}W_t,\qquad v(T)=0.$$

For fixed s and λ, θ_b and $\partial_t\theta_b=s(\alpha_b)_t\theta_b$ are bounded and vanish at both endpoints. Hence the transformed source lies in $L^2_{\mathbb{F}}(a,T;H)$ and $v(T)=0$ in $L^2(\Omega;H)$. By (3.7) and elliptic regularity,

$$\begin{aligned}\mathcal{R}_4(h)&\le C\mathbb{E}\int_{Q_*}\theta_b^2|f|^2\,\mathrm{d}x\,\mathrm{d}t+C\mathbb{E}\int_{Q_*}s^2|(\alpha_b)_t|^2\theta_b^2|h|^2\,\mathrm{d}x\,\mathrm{d}t\\&\le C\mathbb{E}\int_{Q_*}F_s|f|^2\,\mathrm{d}x\,\mathrm{d}t+Cs^{-4}\mathbb{E}\int_{Q_*}B_s^*|h|^2\,\mathrm{d}x\,\mathrm{d}t.\end{aligned}\tag{3.8}$$

The second inequality uses $\xi_b=\xi_*|_{\partial\mathcal{O}}$, $\theta_b^2\le F_s$, $\xi_b\le\xi_*$ and $|(\alpha_b)_t|\le C\xi_b^3$; C includes fixed powers of λ. The right-hand side of (3.7) contains no diffusion norm, so no derivative of g is required.

Step 2: Other boundary terms. Apply the identity on $[a+\varepsilon,T-\varepsilon]$ and let $\varepsilon\downarrow 0$. At fixed cutoff, the continuous H norm controls spatial norms, while the relevant powers of ξ_* times θ_* vanish at a and T. The time-differential terms vanish by domination with the integrable energy supremum. The divergence in [24, Eq. (2.8)] contributes $2\mathbb{E}\int_{\Sigma_*}(V_1+V_2)\cdot\nu\,\mathrm{d}S\,\mathrm{d}t$ to the lower bound. Step 1 controls this contribution without using the Navier reductions in [24, Eqs. (3.4)–(3.7), (3.21)–(3.28)].

For [24, Eqs. (3.8)–(3.13)], integration by parts gives (with summation over repeated indices)

$$\int_{\mathcal{O}} a\, w_{ik}w_{jk}\,\mathrm{d}x = -\int_{\mathcal{O}} a_k w_i w_{jk}\,\mathrm{d}x - \int_{\mathcal{O}} a w_i w_{jkk}\,\mathrm{d}x + \int_{\partial\mathcal{O}} a w_i w_{jk}\nu_k\,\mathrm{d}S,$$
$$\int_{\mathcal{O}} a_{ijkl}w_{ij}w_{kl}\,\mathrm{d}x = \int_{\mathcal{O}} a_{ijkl}w_{il}w_{kj}\,\mathrm{d}x + \int_{\mathcal{O}} \big((a_{ijkl})_l w_i w_{kj} - (a_{ijkl})_j w_i w_{kl}\big)\,\mathrm{d}x + \int_{\partial\mathcal{O}} a_{ijkl}w_i w_{kl}\nu_j\,\mathrm{d}S - \int_{\partial\mathcal{O}} a_{ijkl}w_i w_{kj}\nu_l\,\mathrm{d}S.$$

For smooth a and a_{ijkl}, the boundary terms vanish because $\nabla w = 0$. The first identity applies in [24, Eqs. (3.9), (3.13)], the second in [24, Eqs. (3.11), (3.12)]; no derivatives are transferred in [24, Eqs. (3.8), (3.10)]. The estimate in Step 1 replaces B_2, B_3 and retains the clamped Hessian and third-order traces. The stochastic divergence in [24, Eq. (2.8)], treated in [24, Eqs. (3.14)–(3.15)], vanishes since $w = \nabla w = \mathrm{d}w = \mathrm{d}\nabla w = 0$ at the spectral level.

Step 3: Bulk coercivity. Let $\mathcal{A} = \mathcal{A}_1 + \mathcal{A}_2 + \mathcal{A}_3$, where the nonnegative quantities $\mathcal{A}_j$ bound the interior errors in [24, Eq. (3.2)]. Combining these bounds with [24, Eqs. (3.3), (3.8)–(3.15)], we obtain the analogue of [24, Eq. (3.29)]:

$$2\mathbb{E}\int_{Q_*} \theta_* I f\,\mathrm{d}x\,\mathrm{d}t \geq 2\mathbb{E}\int_{Q_*} I^2\,\mathrm{d}x\,\mathrm{d}t + 2\mathbb{E}\int_{Q_*} I\mathcal{I}_3\,\mathrm{d}x\,\mathrm{d}t + 2\mathcal{K}_s(w) - C\mathcal{A} - C\mathbb{E}\int_{I_*\times\mathcal{O}_\sharp} s^5\lambda^5\xi_*^5|\nabla w|^2\,\mathrm{d}x\,\mathrm{d}t - C\mathbb{E}\int_{Q_*} R_s^*|g|^2\,\mathrm{d}x\,\mathrm{d}t - C_\lambda\mathcal{R}_4(h),$$

where

$$\mathcal{K}_s(w) = \mathbb{E}\int_{Q_*}\big[3s\lambda^2\xi_*|\nabla\Delta w\cdot\nabla\eta|^2 + 32s^3\lambda^4\xi_*^3|D^2 w\nabla\eta\cdot\nabla\eta|^2 + 8s^3\lambda^4\xi_*^3|\nabla\eta|^4|\Delta w|^2 + 22s^5\lambda^6\xi_*^5|\nabla\eta|^4|\nabla w\cdot\nabla\eta|^2 - 16s^5\lambda^6\xi_*^5|\nabla\eta|^6|\nabla w|^2 + 8s^7\lambda^8\xi_*^7|\nabla\eta|^8|w|^2\big]\,\mathrm{d}x\,\mathrm{d}t.$$

The interior coefficients are unchanged. Set

$$m = s^3\lambda^4\xi_*^3|\nabla\eta|^4, \qquad p = s^2\lambda^2\xi_*^2|\nabla\eta|^2 = |\nabla\ell|^2.$$

Since $w = 0$ on $\partial\mathcal{O}$, integration by parts gives

$$\int_{\mathcal{O}}\big(8m|\Delta w|^2 - 16mp|\nabla w|^2 + 8mp^2w^2\big)\,\mathrm{d}x = 8\int_{\mathcal{O}} m(\Delta w + pw)^2\,\mathrm{d}x - 8\int_{\mathcal{O}}\Delta(mp)w^2\,\mathrm{d}x.$$

Since $|\Delta(mp)| \leq Cs^5\lambda^8\xi_*^5$, the last term contributes at most $C\mathcal{A}_1$ after integration and taking expectations. Also,

$$\theta_*\Delta h = (\Delta w + pw) - 2\nabla\ell\cdot\nabla w - (\Delta\ell)w.$$

Apply $(a+b+c)^2 \leq 8a^2 + 2b^2 + 3c^2$. The square in $2\mathcal{K}_s$ gives $16m(\Delta w + pw)^2$; the term with coefficient 44 absorbs $16m|\nabla\ell\cdot\nabla w|^2$, and $m|\Delta\ell|^2|w|^2$ is bounded by $\mathcal{A}_1$. Hence

$$2\mathcal{K}_s(w) \geq 2\mathbb{E}\int_{Q_*} mF_s|\Delta h|^2\,\mathrm{d}x\,\mathrm{d}t - C\mathcal{A}_1.$$

For $\lambda \geq \lambda_0 \geq 1$ and s above the preliminary time-weight threshold, the interior errors satisfy

$$\mathcal{A}_1 \leq C\mathbb{E}\int_{Q_*} (s^5\lambda^8\xi_*^5 + s^6\lambda^7\xi_*^6)|w|^2 \,\mathrm{d}x\,\mathrm{d}t,$$

$$\mathcal{A}_2 \leq C\mathbb{E}\int_{Q_*} (s^3\lambda^6\xi_*^3 + s^4\lambda^5\xi_*^4)|\nabla w|^2 \,\mathrm{d}x\,\mathrm{d}t, \qquad \mathcal{A}_3 \leq C\mathbb{E}\int_{Q_*} (s\lambda^4\xi_* + s^2\lambda^3\xi_*^2)|D^2 w|^2 \,\mathrm{d}x\,\mathrm{d}t.$$

The interior constants depend only on $\mathcal{O}$, T, η and are independent of s, λ; boundary constants are denoted by C_λ. Set $\xi_{\min} = \inf_{(a,T)\times\mathcal{O}} \xi_* > 0$ and require $s\xi_{\min} \geq 1$. The identities $w = \theta_* h$ and $\nabla \ell = s\lambda\xi_* \nabla\eta$ give

$$|\nabla w| \leq C\theta_*(|\nabla h| + s\lambda\xi_*|h|), \qquad |D^2 w| \leq C\theta_*(|D^2 h| + s\lambda\xi_*|\nabla h| + s^2\lambda^2\xi_*^2|h|).$$

The constants are independent of s and λ, since the last term dominates $s\lambda^2\xi_*|h|$ in $D^2 w$ when $s\xi_* \geq 1$. Comparing with $\mathcal{B}_s^*(h)$, we find

$$\mathcal{A} \leq C_A\left(\frac{1}{\lambda} + \frac{1}{s\xi_{\min}}\right)\mathcal{B}_s^*(h),$$

where C_A is independent of both large parameters.

To obtain a Hessian bound uniform in λ, set $\widetilde{h} = s\lambda^2\xi_*\theta_* h$ and let $\mathcal{B}_{0,s}^*(h)$ be the first three terms of $\mathcal{B}_s^*(h)$. The Dirichlet elliptic estimate and product rule, with $s\xi_* \geq 1$ and $\lambda \geq 1$, give

$$\begin{aligned}\mathbb{E}\int_{Q_*} s^2\lambda^4\xi_*^2 F_s|D^2 h|^2 \,\mathrm{d}x\,\mathrm{d}t &\leq C\mathbb{E}\int_{Q_*} |D^2\widetilde{h}|^2 \,\mathrm{d}x\,\mathrm{d}t \\ &\quad + C\mathbb{E}\int_{Q_*} F_s\big(s^4\lambda^6\xi_*^4|\nabla h|^2 + s^6\lambda^8\xi_*^6|h|^2\big)\,\mathrm{d}x\,\mathrm{d}t \leq C\mathcal{B}_{0,s}^*(h).\end{aligned}$$

Here we use $\widetilde{h} \in H^2(\mathcal{O}) \cap H_0^1(\mathcal{O})$ and $\|D^2\widetilde{h}\|_2 \leq C_{\mathcal{O}}\|\Delta\widetilde{h}\|_2$. Expanding $\Delta\widetilde{h}$ gives lower-order terms and $s^2\lambda^4\xi_*^2 F_s|\Delta h|^2$, controlled by $\mathcal{B}_{0,s}^*(h)$.

In the bulk inequality, the bound

$$2\mathbb{E}\int_{Q_*} I\mathcal{I}_3 \,\mathrm{d}x\,\mathrm{d}t \geq -\mathbb{E}\int_{Q_*} I^2 \,\mathrm{d}x\,\mathrm{d}t - C\mathcal{A}$$

from [24, Eq. (3.32)], together with $2\theta_* If \leq I^2 + F_s|f|^2$, cancels the remaining I^2 terms. Combining the Laplacian bound, the positive lower bound for $|\nabla\eta|$ outside $\mathcal{O}_\sharp$ and [24, Remark 2.7], we control $\mathcal{B}_{0,s}^*(h)$, keeping all $\mathcal{A}_j$ on the right. Adding the Hessian bound gives

$$\begin{aligned}\mathcal{B}_s^*(h) \leq C_{\mathrm{int}}\Big[&\mathbb{E}\int_{Q_*} F_s|f|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} R_s^*|g|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathcal{A} \\ &+ \mathbb{E}\int_{I_*\times\mathcal{O}_\sharp} F_s\big(s^7\lambda^8\xi_*^7|h|^2 + s^5\lambda^5\xi_*^5|\nabla h|^2\big)\,\mathrm{d}x\,\mathrm{d}t \\ &+ \mathbb{E}\int_{I_*\times\mathcal{O}_\sharp} F_s s^3\lambda^4\xi_*^3|\Delta h|^2 \,\mathrm{d}x\,\mathrm{d}t\Big] + C_\lambda\mathcal{R}_4(h).\end{aligned}$$

Here C_{int} is independent of s and λ. Choose λ so that $C_{\mathrm{int}}C_A/\lambda \leq 1/4$, then s_0 so that $C_{\mathrm{int}}C_A/(s_0\xi_{\min}) \leq 1/4$, and absorb $C_{\mathrm{int}}\mathcal{A}$. The local observations and $C_\lambda\mathcal{R}_4(h)$ remain. Fix

this λ independently of C_λ. Step 4 will treat the observations and use (3.8) to absorb the remainder.

Step 4: Local estimates and convergence. Choose $\chi \in C_c^\infty(\mathcal{O}_0)$ with $0 \le \chi \le 1$ and $\chi = 1$ on $\mathcal{O}_\sharp$. Set

$$b_\chi = \chi^4 s^3 \lambda^4 \xi_*^3 F_s, \quad \mathcal{L}_\chi = \mathbb{E}\int_{Q_*} b_\chi |\Delta h|^2 \,\mathrm{d}x\,\mathrm{d}t, \quad \mathcal{G}_\chi = \mathbb{E}\int_{Q_*} \chi^4 s^5 \lambda^6 \xi_*^5 F_s |\nabla h|^2 \,\mathrm{d}x\,\mathrm{d}t,$$
$$\mathcal{J}_\chi = \mathbb{E}\int_{Q_{0,*}} O_s^* |h|^2 \,\mathrm{d}x\,\mathrm{d}t, \quad \mathcal{E}_{\mathrm{low}} = \mathbb{E}\int_{Q_*} s^3 \lambda^4 \xi_*^3 F_s |\nabla h|^2 \,\mathrm{d}x\,\mathrm{d}t.$$

With λ fixed, all constants below are uniform in s. Integrating by parts in space, we obtain

$$\mathbb{E}\int_{Q_*} b_\chi h A h \,\mathrm{d}x\,\mathrm{d}t = \mathcal{L}_\chi + 2\mathbb{E}\int_{Q_*} \Delta h \, \nabla b_\chi \cdot \nabla h \,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} \Delta h \, (\Delta b_\chi) h \,\mathrm{d}x\,\mathrm{d}t.$$

The errors are bounded in absolute value by $\mathcal{L}_\chi/2 + C(\mathcal{G}_\chi + \mathcal{J}_\chi + \mathcal{E}_{\mathrm{low}})$, since

$$\frac{|\nabla b_\chi|^2}{b_\chi} \le C F_s \big(\chi^4 s^5 \lambda^6 \xi_*^5 + s^3 \lambda^4 \xi_*^3\big) \mathbf{1}_{\mathcal{O}_0}, \qquad \frac{|\Delta b_\chi|^2}{b_\chi} \le C O_s^* \mathbf{1}_{\mathcal{O}_0}.$$

Set the quotients to zero where $\chi = 0$; differentiating χ^4 gives no negative power of χ. The endpoint terms in Itô's formula vanish, yielding

$$2\mathbb{E}\int_{Q_*} b_\chi h A h \,\mathrm{d}x\,\mathrm{d}t = 2\mathbb{E}\int_{Q_*} b_\chi h f \,\mathrm{d}x\,\mathrm{d}t - \mathbb{E}\int_{Q_*} (b_\chi)_t |h|^2 \,\mathrm{d}x\,\mathrm{d}t - \mathbb{E}\int_{Q_*} b_\chi |g|^2 \,\mathrm{d}x\,\mathrm{d}t.$$

Discard the nonpositive final term. Since $|(b_\chi)_t| + b_\chi^2/F_s \le C O_s^* \mathbf{1}_{\mathcal{O}_0}$,

$$\mathcal{L}_\chi \le C\left(\mathbb{E}\int_{Q_*} F_s |f|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathcal{G}_\chi + \mathcal{J}_\chi + \mathcal{E}_{\mathrm{low}}\right).$$

For $a_\chi = \chi^4 s^5 \lambda^6 \xi_*^5 F_s$, integration by parts gives, for every $\delta > 0$,

$$\mathcal{G}_\chi = -\mathbb{E}\int_{Q_*} a_\chi h \Delta h \,\mathrm{d}x\,\mathrm{d}t + \tfrac{1}{2}\mathbb{E}\int_{Q_*} \Delta a_\chi |h|^2 \,\mathrm{d}x\,\mathrm{d}t \le \delta \mathcal{L}_\chi + C_\delta \mathcal{J}_\chi,$$

Indeed, $a_\chi^2/b_\chi + |\Delta a_\chi| \le C O_s^* \mathbf{1}_{\mathcal{O}_0}$, with zero quotient where $\chi = 0$. Combining these bounds and choosing δ sufficiently small, we absorb $\mathcal{L}_\chi$. Since $\chi = 1$ on $\mathcal{O}_\sharp$ and $\lambda \ge 1$, this controls both derivative observations in Step 3. The lower-order term satisfies $\mathcal{E}_{\mathrm{low}} \le (s\lambda^2 \xi_{\min})^{-1} \mathcal{B}_s^*(h)$ and is absorbed by increasing s_0. Hence

$$\mathcal{B}_s^*(h) \le C\left[\mathbb{E}\int_{Q_*} F_s |f|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} R_s^* |g|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_{0,*}} O_s^* |h|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathcal{R}_4(h)\right].$$

Increase s_0 once more and use (3.8) to absorb the remainder, obtaining (3.6) at the spectral level.

With s and λ fixed, we remove the cutoff using (2.7): states converge in $L^2_{\mathbb{F}}(a, T; V)$, sources and diffusions converge in $L^2_{\mathbb{F}}(a, T; H)$, and weights are bounded. The positive Hessian term in (3.4) has been discarded, so no spatial regularity of g is needed. Take $h = z$, $f = -\Phi$ and $g = -Z$. □

3.3. **The adjoint equation with distributional sources.** Let Φ_0, Φ_1, Φ_2 be progressively measurable, with values in $L^2(\mathcal{O})$, $L^2(\mathcal{O};\mathbb{R}^n)$ and $L^2(\mathcal{O};\mathbb{R}^{n\times n})$, respectively, and square integrable on $\Omega\times(0,T)$. Define

$$\langle \mathfrak{D}\Phi, \psi\rangle_{V',V} = \int_{\mathcal{O}} \big(\Phi_0\psi - \Phi_1\cdot\nabla\psi + \Phi_2 : D^2\psi\big)\,\mathrm{d}x, \qquad \psi\in V. \tag{3.9}$$

The interior distribution $\mathfrak{D}\Phi = \Phi_0 + \operatorname{div}\Phi_1 + \sum_{i,j}\partial_{ij}(\Phi_2)_{ij}$ is extended to V' by (3.9). We consider

$$\begin{cases} \mathrm{d}z - Az\,\mathrm{d}t = \mathfrak{D}\Phi\,\mathrm{d}t + Z\,\mathrm{d}W_t, \\ \qquad z(T) = z_T \in L^2(\Omega,\mathcal{F}_T;H). \end{cases} \tag{3.10}$$

Lemma 3.4. *Let $A\in\{A_N, A_C\}$, and let (Φ_0,Φ_1,Φ_2) satisfy the preceding measurability and integrability conditions. Define $\mathfrak{D}\Phi$ by (3.9). Then, for every $z_T\in L^2(\Omega,\mathcal{F}_T;H)$, equation (3.10) has a unique variational solution $(z,Z)\in\mathcal{H}\times L^2_{\mathbb{F}}(0,T;H)$ satisfying*

$$\|z\|^2_{\mathcal{H}} + \|Z\|^2_{L^2_{\mathbb{F}}(0,T;H)} \le C\left(\mathbb{E}\|z_T\|^2_H + \sum_{j=0}^{2}\|\Phi_j\|^2_{L^2_{\mathbb{F}}(0,T;L^2)}\right). \tag{3.11}$$

Here $C>0$ depends only on $\mathcal{O}$, T, and the boundary realization. Spatial L^2 norms are over $\mathcal{O}$, with scalar, vector or matrix values as appropriate.

For every forward solution Y in Lemma 2.3, the duality identity reads

$$\begin{aligned} \mathbb{E}(Y(T),z_T)_H &= \mathbb{E}(Y(0),z(0))_H + \mathbb{E}\int_{Q_0} uz\,\mathrm{d}x\,\mathrm{d}t \\ &\quad + \mathbb{E}\int_Q \big(\phi z + Y\Phi_0 - \nabla Y\cdot\Phi_1 + D^2Y : \Phi_2 + UZ\big)\,\mathrm{d}x\,\mathrm{d}t. \end{aligned} \tag{3.12}$$

The conclusions hold on $[a,T]$ with the restricted filtration and $\mathcal{F}_a$ as initial σ-field.

Proof. Step 1: The source in V'. By $V\hookrightarrow H^2(\mathcal{O})$ and Cauchy–Schwarz in (3.9),

$$\|\mathfrak{D}\Phi\|^2_{V'} \le C_{\mathcal{O}}\big(\|\Phi_0\|^2_{L^2(\mathcal{O})} + \|\Phi_1\|^2_{L^2(\mathcal{O};\mathbb{R}^n)} + \|\Phi_2\|^2_{L^2(\mathcal{O};\mathbb{R}^{n\times n})}\big).$$

Since the bounded linear map $(\Phi_0,\Phi_1,\Phi_2)\mapsto\mathfrak{D}\Phi$ preserves progressive measurability, we have $\mathfrak{D}\Phi\in L^2_{\mathbb{F}}(0,T;V')$.

Step 2: Galerkin estimates. Let (e_k) be an orthonormal eigenbasis of A, with spectral projection P_N on H. For $\Psi\in V'$, set

$$P_N^{V'}\Psi = \sum_{k=1}^{N}\langle\Psi,e_k\rangle_{V',V}e_k.$$

These projections contract the dual norm induced by $\|\cdot\|_V$ and converge strongly on V'. Lemma 2.3, with source $P_N^{V'}\mathfrak{D}\Phi$ and terminal datum P_Nz_T, gives (z_N,Z_N). Itô's formula from t to T, followed by expectation, gives

$$\mathbb{E}\|z_N(t)\|^2_H + \mathbb{E}\int_t^T\big(2\|z_N\|^2_V + \|Z_N\|^2_H\big)\,\mathrm{d}\tau = \mathbb{E}\|P_Nz_T\|^2_H - 2\mathbb{E}\int_t^T\langle\mathfrak{D}\Phi,z_N\rangle\,\mathrm{d}\tau.$$

The source pairing satisfies

$$2|\langle \mathfrak{D}\Phi, z_N\rangle| \le \|z_N\|_V^2 + \|\mathfrak{D}\Phi\|_{V'}^2.$$

Absorption yields uniform mean-energy bounds. Step 2 of Lemma 2.3 then applies: the localized BDG estimate uses only state and diffusion norms, with the source pairing controlled above. This gives the $\mathcal{H} \times L^2_{\mathbb{F}}(0,T;H)$ bound with source norm in $L^2_{\mathbb{F}}(0,T;V')$.

Applying the difference estimate to the convergent projected data gives strong convergence of (z_N, Z_N) in $\mathcal{H} \times L^2_{\mathbb{F}}(0,T;H)$. Continuity of $A : V \to V'$ and of the stochastic integral yields the limiting equation; Step 1 gives (3.11); uniqueness follows from the difference estimate.

Step 3: Duality. For $Y_N = P_N Y$, repeat Step 3 of Lemma 2.3 at the spectral level: form cancellation, covariation and zero-mean martingales are unchanged. For the new source pairing,

$$\mathbb{E}\int_0^T |\langle \mathfrak{D}\Phi, Y_N - Y\rangle|\,\mathrm{d}t \le \|\mathfrak{D}\Phi\|_{L^2_{\mathbb{F}}(0,T;V')}\|Y_N - Y\|_{L^2_{\mathbb{F}}(0,T;V)} \longrightarrow 0.$$

Strong L^2 convergence justifies the limit in the other pairings. Expanding (3.9) gives (3.12) for (2.8). On $[a,T]$, use $W_{a+t} - W_a$ and the shifted filtration in Subsection 2.3. □

Remark 3.5. For clamped conditions, $V' = H^{-2}(\mathcal{O})$; for Navier conditions, $V' = (H^2(\mathcal{O}) \cap H_0^1(\mathcal{O}))'$, which includes the boundary contribution of Φ_2. For smooth z and Φ_2 up to the boundary, and $\psi \in H^2(\mathcal{O}) \cap H_0^1(\mathcal{O})$, Green's formula gives

$$a(z,\psi) = \int_{\mathcal{O}} \Delta^2 z\,\psi\,\mathrm{d}x + \int_{\partial\mathcal{O}} \Delta z\,\partial_\nu\psi\,\mathrm{d}S,$$

$$\sum_{i,j}\int_{\mathcal{O}} (\Phi_2)_{ij}\partial_{ij}\psi\,\mathrm{d}x = \sum_{i,j}\int_{\mathcal{O}} \partial_{ij}(\Phi_2)_{ij}\,\psi\,\mathrm{d}x + \int_{\partial\mathcal{O}} (\nu^\top\Phi_2\nu)\,\partial_\nu\psi\,\mathrm{d}S.$$

Since $\nabla\psi = \nu\partial_\nu\psi$ and ψ vanishes on the boundary, the boundary coefficient in $\mathrm{d}z - Az\,\mathrm{d}t = \mathfrak{D}\Phi\,\mathrm{d}t + Z\,\mathrm{d}W_t$ is $-\Delta z - \nu^\top\Phi_2\nu$. Varying $\partial_\nu\psi$, we obtain

$$\Delta z = -\nu^\top\Phi_2\nu \quad \text{on } \partial\mathcal{O}.$$

For clamped tests, both boundary integrals vanish. For L^2 tensors, (3.9) encodes the relation without a trace of Φ_2. The controlled forward equation has an H-valued drift and retains homogeneous Navier conditions.

3.4. **An auxiliary null-control problem.** On $I_* = (a,T)$, we use (3.1)–(3.2), with $\sigma = s\lambda\xi_*$ and fixed λ. Increase s_0 to ensure $\sigma \ge 1$. In this subsection, all spaces and integrals are over I_*, with constants uniform in $s \ge s_0$. Thus

$$B_s^* = \lambda^2 F_s\sigma^6, \qquad O_s^* = \lambda F_s\sigma^7, \qquad R_s^* = e^{C_*\lambda M}F_s\sigma^4.$$

Lemma 3.6. *There exists a constant $C > 0$ such that, for every $s \ge s_0$ and every progressively measurable source ϕ satisfying*

$$\mathbb{E}\int_{Q_*} (B_s^*)^{-1}|\phi|^2\,\mathrm{d}x\,\mathrm{d}t < \infty,$$

there are controls $u \in L^2_{\mathbb{F}}(a,T;H_0)$, $U \in L^2_{\mathbb{F}}(a,T;H)$, for which the associated variational solution $Y \in L^2_{\mathbb{F}}(\Omega;C([a,T];H)) \cap L^2_{\mathbb{F}}(a,T;V)$ satisfies

$$\begin{cases} \mathrm{d}Y + AY\,\mathrm{d}t = (\phi + Bu)\,\mathrm{d}t + U\,\mathrm{d}W_t \quad \text{on } I_*, \\ \qquad Y(a) = 0, \qquad Y(T) = 0. \end{cases}$$

The endpoint identities hold in $L^2(\Omega;H)$, and

$$\begin{aligned} \mathcal{C}_*(Y,u,U) &:= \mathbb{E}\int_{Q_*} F_s^{-1}|Y|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_{0,*}} (O_s^*)^{-1}|u|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} (R_s^*)^{-1}|U|^2\,\mathrm{d}x\,\mathrm{d}t \\ &\le C\mathbb{E}\int_{Q_*} (B_s^*)^{-1}|\phi|^2\,\mathrm{d}x\,\mathrm{d}t. \end{aligned} \tag{3.13}$$

For fixed s, the triple (Y,u,U) can be chosen linearly in ϕ. Here C depends only on $\mathcal{O}$, $\mathcal{O}_0$, T, the fixed weight and boundary realization, independently of s and ϕ.

Proof. Fix $s \ge s_0$.

Step 1: Observation space. Let (p,P) solve (3.5) with source $\Psi \in L^2_{\mathbb{F}}(a,T;H)$ and terminal datum $p_T \in L^2(\Omega,\mathcal{F}_T;H)$. By (3.6),

$$\begin{aligned} \mathbb{E}\int_{Q_*} B_s^*|p|^2\,\mathrm{d}x\,\mathrm{d}t \le C_0\Big(&\mathbb{E}\int_{Q_{0,*}} O_s^*|p|^2\,\mathrm{d}x\,\mathrm{d}t \\ &+ \mathbb{E}\int_{Q_*} F_s|\Psi|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} R_s^*|P|^2\,\mathrm{d}x\,\mathrm{d}t\Big). \end{aligned} \tag{3.14}$$

Let $\mathfrak{E}_*$ be the closure of the triples $e = (p|_{\mathcal{O}_0}, \Psi, P)$ in the product of the corresponding progressive weighted L^2 spaces. For $e = (\zeta,\Psi,P)$ and $\widetilde{e} = (\widetilde{\zeta},\widetilde{\Psi},\widetilde{P})$, set

$$\mathfrak{a}_*(e,\widetilde{e}) = \mathbb{E}\int_{Q_{0,*}} O_s^*\zeta\widetilde{\zeta}\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} F_s\Psi\widetilde{\Psi}\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} R_s^*P\widetilde{P}\,\mathrm{d}x\,\mathrm{d}t.$$

Positivity of the weights on I_* makes the product and its closed subspace $\mathfrak{E}_*$ Hilbert spaces. By (3.14), $e \mapsto p$ defines a bounded map into $L^2(B_s^*)$ and extends linearly to $\mathfrak{E}_*$. By weighted Cauchy–Schwarz, we extend the functional

$$\mathcal{L}_*(e) = \mathbb{E}\int_{Q_*} \phi p\,\mathrm{d}x\,\mathrm{d}t,$$

with

$$|\mathcal{L}_*(e)|^2 \le C_0\left(\mathbb{E}\int_{Q_*} (B_s^*)^{-1}|\phi|^2\,\mathrm{d}x\,\mathrm{d}t\right)\mathfrak{a}_*(e,e).$$

Step 2: Riesz controls. The Riesz representation theorem gives a unique $\widehat{e} = (\widehat{\zeta},\widehat{\Psi},\widehat{P}) \in \mathfrak{E}_*$ such that

$$\mathfrak{a}_*(\widehat{e},e) = \mathcal{L}_*(e), \qquad e \in \mathfrak{E}_*.$$

Set

$$Y_c = -F_s\widehat{\Psi}, \qquad u = -O_s^*\widehat{\zeta}, \qquad U = -R_s^*\widehat{P}.$$

These processes are progressive. From the Riesz bound, we obtain

$$\mathbb{E}\int_{Q_*} F_s^{-1}|Y_c|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_{0,*}} (O_s^*)^{-1}|u|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} (R_s^*)^{-1}|U|^2\,\mathrm{d}x\,\mathrm{d}t = \|\mathcal{L}_*\|^2_{\mathfrak{E}_*'}$$

$$\leq C_0\mathbb{E}\int_{Q_*} (B_s^*)^{-1}|\phi|^2\,\mathrm{d}x\,\mathrm{d}t. \tag{3.15}$$

For fixed s, boundedness of F_s, O_s^* and R_s^* gives unweighted L^2 integrability of Y_c, u, U; boundedness of B_s^* gives $\phi \in L^2_{\mathbb{F}}(a,T;H)$.

Step 3: State identification. Let Y solve variationally with these controls and $Y(a)=0$. For admissible (p_T,Ψ), duality and the Riesz identity give

$$\mathbb{E}(Y(T),p_T)_H = \mathcal{L}_*(e) + \mathbb{E}\int_{Q_{0,*}} up\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*}(UP+Y\Psi)\,\mathrm{d}x\,\mathrm{d}t = \mathbb{E}\int_{Q_*}(Y-Y_c)\Psi\,\mathrm{d}x\,\mathrm{d}t.$$

Taking $p_T=0$ and $\Psi = Y-Y_c$ is admissible by the unweighted L^2 bounds and gives $Y=Y_c$ in $L^2_{\mathbb{F}}(a,T;H)$. Taking $\Psi=0$ and $p_T=Y(T)$ then gives $\mathbb{E}\|Y(T)\|_H^2=0$.

From (3.15), we have obtained (3.13). For fixed s, the Riesz and forward maps make the selection (Y,u,U) linear in ϕ. The cost depends only on C_0 in (3.14) and is uniform in s. □

3.5. **Weighted regularity of the auxiliary trajectory.** We next estimate the auxiliary trajectory's gradient and Hessian without differentiating the diffusion. See [17, Lemma 2.8] and [19] for deterministic controls, and [33, Lemma 2.2] for one-dimensional stochastic controls. Here the bound is multidimensional and uniform in the Carleman parameter.

Lemma 3.7. *For (Y,u,U) from Lemma 3.6, there exists $C>0$, depending only on the quantities in that lemma, such that, for every $s\geq s_0$,*

$$\mathcal{C}_*(Y,u,U) + \mathbb{E}\int_{Q_*} F_s^{-1}\big(\sigma^{-2}|\nabla Y|^2 + \sigma^{-4}|D^2Y|^2\big)\,\mathrm{d}x\,\mathrm{d}t \leq C\mathbb{E}\int_{Q_*}(B_s^*)^{-1}|\phi|^2\,\mathrm{d}x\,\mathrm{d}t. \tag{3.16}$$

The constant is independent of s and ϕ. For U, only the weighted L^2 bound in (3.13) is required.

Proof. Set $\mathcal{P}_\phi = \mathbb{E}\int_{Q_*}(B_s^*)^{-1}|\phi|^2\,\mathrm{d}x\,\mathrm{d}t$. Constants depend only on $\mathcal{O},T,\eta$ and the fixed λ, and are uniform in $s\geq s_0$ and the regularization index.

Step 1: Multipliers. Write $d(t)=\sqrt{(t-a)(T-t)}$, $E=e^{\lambda(2M+\eta)}$ and $K=e^{4\lambda M}$, so that $\xi_*=E/d$ and $\alpha_*=(E-K)/d$. Differentiating with $\beta=-d'/d$, we find

$$(\xi_*)_t=\beta\xi_*,\quad (\alpha_*)_t=\beta\alpha_*,\quad |\beta|\leq C\xi_*^2,\quad |(\xi_*)_t|+|(\alpha_*)_t|\leq C\xi_*^3.$$

For $m=F_s^{-1}\sigma^{-4}$ and $\rho=\log m$, we have

$$\rho_i=-2(\sigma+2\lambda)\eta_i,\qquad \rho_t=-\beta(2s\alpha_*+4),\qquad \rho_{ij}=-2(\sigma+2\lambda)\eta_{ij}-2\lambda\sigma\eta_i\eta_j.$$

Since $\sigma\geq 1$ and λ is fixed, we obtain $|\nabla\rho|\leq C\sigma$, $|D^2\rho|\leq C\sigma^2$ and $|\rho_t|\leq C(s\xi_*^3+\xi_*^2)\leq C\sigma^4$. Regularize m at the endpoints by

$$m_N=\frac{m}{1+m/N}=\frac{N}{1+NF_s\sigma^4},\qquad r_N=\frac{N}{N+m},\qquad N\geq 1.$$

Since $m_N = mr_N$ and $(r_N)_i = -r_N(1-r_N)\rho_i$,

$$(m_N)_i = m_N r_N \rho_i, \qquad (m_N)_t = m_N r_N \rho_t, \qquad (m_N)_{ij} = m_N\big[r_N\rho_{ij} + r_N(2r_N-1)\rho_i\rho_j\big].$$

Uniformly for $N \geq 1$, $s \geq s_0$,

$$|\nabla m_N| \leq K_1 m_N \sigma, \quad |D^2 m_N| + |\Delta m_N| \leq K_2 m_N \sigma^2, \quad |(m_N)_t| \leq K_t m_N \sigma^4.$$

For fixed s, $F_s\sigma^4 = (s\lambda)^4 E^4 d^{-4} e^{-2s(K-E)/d}$ and the required derivatives vanish at a, T, since $K - E$ is bounded away from zero. Thus m_N extends as N at the endpoints; for fixed N, s, its first time and second spatial derivatives are bounded.

Step 2: Spatial estimates. Fix $t \in I_*$ and $v \in V$. Set

$$L_N(v) = \int_{\mathcal{O}} m_N |\Delta v|^2 \,\mathrm{d}x, \qquad G_N(v) = \int_{\mathcal{O}} m_N \sigma^2 |\nabla v|^2 \,\mathrm{d}x, \qquad K_N(v) = \int_{\mathcal{O}} m_N \sigma^4 |v|^2 \,\mathrm{d}x.$$

Suppress the argument v and set $a_N = m_N\sigma^2$. Step 1 and $\nabla\sigma = \lambda\sigma\nabla\eta$ give $|\Delta a_N| \leq K_a m_N \sigma^4$. For smooth v with the essential boundary conditions, integration by parts gives

$$G_N = -\int_{\mathcal{O}} a_N v \Delta v \,\mathrm{d}x + \frac{1}{2}\int_{\mathcal{O}} \Delta a_N |v|^2 \,\mathrm{d}x \leq \epsilon L_N + \left(\frac{1}{4\epsilon} + \frac{K_a}{2}\right) K_N, \qquad \epsilon > 0. \tag{3.17}$$

The zero Dirichlet trace removes the boundary terms. Density then extends the bound to $v \in V$. Here $a_N^2/m_N = m_N\sigma^4$. Multiplication by m_N preserves V, and

$$a(v, m_N v) = L_N + 2\int_{\mathcal{O}} \Delta v \,\nabla m_N \cdot \nabla v \,\mathrm{d}x + \int_{\mathcal{O}} \Delta v \,\Delta m_N \, v \,\mathrm{d}x.$$

Young's inequality gives the respective error bounds $L_N/8 + 8K_1^2 G_N$ and $L_N/8 + 2K_2^2 K_N$. It follows from (3.17), with $\epsilon = [4(1+8K_1^2)]^{-1}$, that

$$a(v, m_N v) \geq \tfrac{1}{2} L_N - C_a K_N. \tag{3.18}$$

To estimate the Hessian, set $q_N = \sqrt{m_N}$ and $w = q_N v$. Step 1 gives $|\nabla q_N| \leq C q_N \sigma$ and $|D^2 q_N| + |\Delta q_N| \leq C q_N \sigma^2$, while

$$\Delta w = q_N \Delta v + 2\nabla q_N \cdot \nabla v + (\Delta q_N) v, \qquad q_N v_{ij} = w_{ij} - (q_N)_i v_j - (q_N)_j v_i - (q_N)_{ij} v.$$

Since $w \in H^2(\mathcal{O}) \cap H_0^1(\mathcal{O})$, elliptic regularity gives $\|D^2 w\|_2^2 \leq C_{\mathcal{O}} \|\Delta w\|_2^2$. Hence, using (3.17),

$$\begin{aligned} \|\Delta w\|_2^2 &\leq C(L_N + G_N + K_N) \leq C(L_N + K_N), \\ \int_{\mathcal{O}} m_N |D^2 v|^2 \,\mathrm{d}x &\leq 4\|D^2 w\|_2^2 + 8\int_{\mathcal{O}} |\nabla q_N|^2 |\nabla v|^2 \,\mathrm{d}x + 4\int_{\mathcal{O}} |D^2 q_N|^2 |v|^2 \,\mathrm{d}x \\ &\leq C(L_N + K_N). \end{aligned} \tag{3.19}$$

Both realizations satisfy $V \subset H^2(\mathcal{O}) \cap H_0^1(\mathcal{O})$.

Step 3: Itô's identity. For fixed N, s, multiplication by m_N is bounded on H and V, uniformly in time. It preserves the clamped normal trace since $\partial_\nu(m_N v) = m_N \partial_\nu v + v\partial_\nu m_N = 0$; Navier tests require only zero Dirichlet trace. The domain $D(A_N)$ need not be preserved:

for smooth $v \in D(A_N)$, $\Delta(m_N v)|_{\partial\mathcal{O}} = 2\partial_\nu m_N \partial_\nu v$. We therefore use the form pairing in the weighted energy identity:

$$\begin{aligned} \mathrm{d}\int_{\mathcal{O}} m_N|Y|^2\,\mathrm{d}x = \Big[\int_{\mathcal{O}} (m_N)_t|Y|^2\,\mathrm{d}x - 2a(Y, m_N Y) \\ + 2\int_{\mathcal{O}} m_N Y(\phi + Bu)\,\mathrm{d}x + \int_{\mathcal{O}} m_N|U|^2\,\mathrm{d}x\Big]\mathrm{d}t + 2(m_N Y, U)_H\,\mathrm{d}W_t. \end{aligned} \tag{3.20}$$

The correction $\int_{\mathcal{O}} m_N|U|^2\,\mathrm{d}x$ requires no spatial derivative of U.

For the spectral projections P_k of A, set $Y_k = P_k Y$, $b_k = P_k(\phi + Bu)$, $U_k = P_k U$. Apply finite-dimensional Itô's formula with the multiplier and BDG as in Lemma 2.3. For fixed N, $Y_k \to Y$ in $\mathcal{H}$ and $(b_k, U_k) \to (\phi + Bu, U)$ in $L^2(H)^2$. Bounded multiplication on V justifies the limit in the drift pairings without commutation; boundedness on the pivot space gives martingale convergence in expected supremum norm. Thus (3.20) holds; justification constants may depend on N, s. Taking expectations on $[a, T]$ and using the zero endpoint values of Y yields

$$\begin{aligned} 2\mathbb{E}\int_a^T a(Y, m_N Y)\,\mathrm{d}t = \mathbb{E}\int_{Q_*} \big((m_N)_t|Y|^2 + 2m_N Y\phi + m_N|U|^2\big)\,\mathrm{d}x\,\mathrm{d}t \\ + 2\mathbb{E}\int_{Q_{0,*}} m_N Y u\,\mathrm{d}x\,\mathrm{d}t. \end{aligned} \tag{3.21}$$

Step 4: Costs and convergence. Write the auxiliary costs as

$$\mathcal{C}_y = \mathbb{E}\int_{Q_*} F_s^{-1}|Y|^2\,\mathrm{d}x\,\mathrm{d}t, \quad \mathcal{C}_u = \mathbb{E}\int_{Q_{0,*}} (O_s^*)^{-1}|u|^2\,\mathrm{d}x\,\mathrm{d}t, \quad \mathcal{C}_U = \mathbb{E}\int_{Q_*} (R_s^*)^{-1}|U|^2\,\mathrm{d}x\,\mathrm{d}t.$$

The auxiliary control estimate gives $\mathcal{C}_y + \mathcal{C}_u + \mathcal{C}_U \le C_H \mathcal{P}_\phi$. In (3.21), we use $m_N \le F_s^{-1}\sigma^{-4}$ and

$$m_N\sigma^4 \le F_s^{-1}, \qquad m_N R_s^* \le e^{C_*\lambda M}, \qquad m_N^2 F_s B_s^* \le \lambda^2\sigma^{-2}, \qquad m_N^2 F_s O_s^* \le \lambda\sigma^{-1}.$$

Choose uniform upper bounds K_ϕ and K_u for $\lambda^2\sigma^{-2}$ and $\lambda\sigma^{-1}$ on Q_* for $s \ge s_0$, and set $K_U = e^{C_*\lambda M}$; $\inf_{Q_*}\sigma$ is uniformly bounded below. Young's inequality gives

$$\mathbb{E}\int_{Q_*} |(m_N)_t||Y|^2\,\mathrm{d}x\,\mathrm{d}t + 2C_a\mathbb{E}\int_a^T K_N(Y)\,\mathrm{d}t \le (K_t + 2C_a)\mathcal{C}_y,$$

$$2\mathbb{E}\int_{Q_*} m_N|Y\phi|\,\mathrm{d}x\,\mathrm{d}t \le \mathcal{C}_y + K_\phi\mathcal{P}_\phi,$$

$$2\mathbb{E}\int_{Q_{0,*}} m_N|Yu|\,\mathrm{d}x\,\mathrm{d}t \le \mathcal{C}_y + K_u\mathcal{C}_u, \qquad \mathbb{E}\int_{Q_*} m_N|U|^2\,\mathrm{d}x\,\mathrm{d}t \le K_U\mathcal{C}_U.$$

By (3.21) and (3.18), $\mathbb{E}\int_a^T L_N(Y)\,\mathrm{d}t \le C\mathcal{P}_\phi$. Together with (3.17) and (3.19), this yields

$$\mathbb{E}\int_{Q_*} m_N(|\Delta Y|^2 + \sigma^2|\nabla Y|^2 + |D^2 Y|^2)\,\mathrm{d}x\,\mathrm{d}t \le C\mathcal{P}_\phi.$$

The bound is uniform in N and s. After passing to the spectral limit at fixed N, let $N \to \infty$; $m_N \uparrow F_s^{-1}\sigma^{-4}$ and monotone convergence give

$$\mathbb{E}\int_{Q_*} F_s^{-1}\big(\sigma^{-4}|\Delta Y|^2 + \sigma^{-2}|\nabla Y|^2 + \sigma^{-4}|D^2 Y|^2\big)\,\mathrm{d}x\,\mathrm{d}t \le C\mathcal{P}_\phi.$$

Adding (3.13) and discarding the Laplacian term, we obtain (3.16), uniformly in $s \ge s_0$. □

3.6. **The Carleman estimate with distributional sources.** The weighted auxiliary bounds and duality yield a Carleman estimate for distributional sources.

Theorem 3.8. *Let $A \in \{A_N, A_C\}$ and fix λ as in Proposition 3.2. There exist constants $s_0, C > 0$ such that, for every $s \ge s_0$, every $z_T \in L^2(\Omega, \mathcal{F}_T; H)$ and every source triple satisfying the hypotheses of Lemma 3.4 on $I_* = (a, T)$, the corresponding variational solution (z, Z) satisfies*

$$\begin{aligned}\mathbb{E}\int_{Q_*} B_s^*|z|^2\,\mathrm{d}x\,\mathrm{d}t \le C\Big[&\mathbb{E}\int_{Q_{0,*}} O_s^*|z|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} R_s^*|Z|^2\,\mathrm{d}x\,\mathrm{d}t\\ &+ \mathbb{E}\int_{Q_*} F_s\big(|\Phi_0|^2 + \sigma^2|\Phi_1|^2 + \sigma^4|\Phi_2|^2\big)\,\mathrm{d}x\,\mathrm{d}t\Big].\end{aligned} \tag{3.22}$$

Here $\sigma = s\lambda\xi_*$. Constants depend only on $\mathcal{O}, \mathcal{O}_0, T$, the fixed weight and boundary realization, and are independent of s and the data. The source in V' is defined by (3.9), with no boundary trace of Φ_2.

Proof. Fix $s \ge s_0$.

Step 1: Auxiliary source. Set

$$\mathcal{D} = \mathbb{E}\int_{Q_*} B_s^*|z|^2\,\mathrm{d}x\,\mathrm{d}t, \qquad \phi = B_s^* z.$$

The energy bound for z and boundedness of B_s^* give $\mathcal{D} < \infty$ and $\phi \in L^2_{\mathbb{F}}(a, T; H)$. Since $B_s^* > 0$ on $I_* \times \mathcal{O}$, $\mathbb{E}\int_{Q_*}(B_s^*)^{-1}|\phi|^2\,\mathrm{d}x\,\mathrm{d}t = \mathcal{D}$. Lemmas 3.6 and 3.7 give (Y, u, U) with $Y(a) = Y(T) = 0$ in $L^2(\Omega; H)$ and

$$\begin{aligned}\mathcal{A} = \mathbb{E}\int_{Q_*} F_s^{-1}(|Y|^2 + \sigma^{-2}|\nabla Y|^2 + \sigma^{-4}|D^2 Y|^2)\,\mathrm{d}x\,\mathrm{d}t\\ + \mathbb{E}\int_{Q_{0,*}} (O_s^*)^{-1}|u|^2\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} (R_s^*)^{-1}|U|^2\,\mathrm{d}x\,\mathrm{d}t \le C_A\mathcal{D}.\end{aligned}$$

Step 2: Duality. In (3.12) on I_*, the endpoint terms vanish; the definition of ϕ gives

$$0 = \mathcal{D} + \mathbb{E}\int_{Q_{0,*}} uz\,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_{Q_*} (UZ + Y\Phi_0 - \nabla Y\cdot\Phi_1 + D^2 Y : \Phi_2)\,\mathrm{d}x\,\mathrm{d}t.$$

The source pairings define $\mathfrak{D}\Phi$; no integration by parts or traces of Φ_1 or Φ_2 are used.

Step 3: Weighted estimate. Let $\mathcal{R}$ be the right-hand bracket in (3.22). By weighted Cauchy–Schwarz in Step 2, with $\mathcal{A}$ and the Frobenius matrix inner product, we have

$$\mathcal{D} \le \mathcal{A}^{1/2}\mathcal{R}^{1/2} \le (C_A\mathcal{D})^{1/2}\mathcal{R}^{1/2} \le \tfrac{1}{2}\mathcal{D} + \tfrac{1}{2}C_A\mathcal{R}.$$

After absorbing $\mathcal{D}/2$, we obtain $\mathcal{D} \le C_A \mathcal{R}$, including $\mathcal{D} = 0$, uniformly in s. □

Remark 3.9. The source weights have the same orders as those in [17, Proposition 2.4]. The Itô correction $m_N|U|^2$ is controlled by the diffusion cost $(R_s^*)^{-1}|U|^2$.

4. Linear null controllability

4.1. **Temporal weights.** In Sections 4–6, we omit the Carleman parameter from subscripts; all weights, spaces and control operators depend on both parameters. Choose s and κ independently. Let $\zeta \in C^\infty([0,T];[0,1])$ be nonincreasing, equal to one on $[0,b]$ and zero on $[c,T]$, and set $L_\zeta = \|\zeta'\|_\infty$, $m_s = \min_{J_t \times \overline{\mathcal{O}}} B_s^* > 0$. For $s \ge s_0$ and $\kappa \ge 1$, define

$$W_\kappa(t) = \frac{m_s}{4(1+L_\zeta)} e^{-\kappa(c-t)}\zeta(t), \qquad \varepsilon_\kappa = \frac{m_s}{4(1+L_\zeta)} e^{-\kappa c}, \tag{4.1}$$

$$b_\kappa = B_s^* + \kappa W_\kappa, \qquad o_\kappa = O_s^* + W_\kappa,$$

$$r_\kappa = R_s^* + W_\kappa, \qquad q_{j,\kappa} = F_s \sigma^{2j} + \kappa^{-1+j/2} W_\kappa, \quad j = 0,1,2. \tag{4.2}$$

Terms containing F_s are extended by zero to $t \le a$; $b = T/2$ denotes the cutoff time, whereas b_κ is a weight. Set

$$\mathcal{S}_\kappa = L^2_{\mathcal{P}}(\Omega \times Q; b_\kappa^{-1}), \qquad \|\phi\|^2_{\mathcal{S}_\kappa} = \mathbb{E}\int_Q b_\kappa^{-1}|\phi|^2 \,\mathrm{d}x\,\mathrm{d}t.$$

Multiplication by $\sqrt{w}$ maps the progressive L^2 space with a positive deterministic weight w isometrically onto its unweighted counterpart. The weighted spaces are Hilbert spaces even for unbounded w or w^{-1}. Boundedness of b_κ gives $\mathcal{S}_\kappa \hookrightarrow L^2_{\mathbb{F}}(0,T;H)$. Vector and matrix norms are taken componentwise.

4.2. **The null-control operator.**

Proposition 4.1. *Let $A \in \{A_N, A_C\}$. There exist $C_R, c_0, c_1, c_2, C > 0$, depending only on $\mathcal{O}, \mathcal{O}_0, T$, the fixed spatial weight and the boundary realization, such that, for $s \ge s_0$ and $\kappa \ge 1$, the following hold.*

(i) The weights $b_\kappa, o_\kappa, r_\kappa$ and $q_{j,\kappa}$ are positive almost everywhere on Q and at most C, uniformly in s, κ. Moreover,

$$\frac{q_{j,\kappa}}{b_\kappa} \le \rho_j(s,\kappa) := c_j s^{-6+2j} + \kappa^{-2+j/2}, \qquad j = 0,1,2, \tag{4.3}$$

$$\varepsilon_\kappa^{-1} \le C e^{C(s+\kappa)}. \tag{4.4}$$

(ii) Every solution (z,Z) given by Lemma 3.4 satisfies

$$\varepsilon_\kappa \mathbb{E}\|z(0)\|_H^2 + \mathbb{E}\int_Q b_\kappa |z|^2 \,\mathrm{d}x\,\mathrm{d}t$$

$$\le C_R\Big[\mathbb{E}\int_{Q_0} o_\kappa |z|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathbb{E}\int_Q r_\kappa |Z|^2 \,\mathrm{d}x\,\mathrm{d}t + \sum_{j=0}^{2} \mathbb{E}\int_Q q_{j,\kappa}|\Phi_j|^2 \,\mathrm{d}x\,\mathrm{d}t\Big]. \tag{4.5}$$

(iii) There exists a bounded linear null-control operator

$$\mathcal{R}_\kappa : L^2(\Omega, \mathcal{F}_0; H) \times \mathcal{S}_\kappa \longrightarrow \mathcal{H} \times \mathcal{U}.$$

For $(y, u, U) = \mathcal{R}_\kappa(y_0, \phi)$, the state solves (2.8) in the variational sense, $y(T) = 0$ in $L^2(\Omega; H)$, and

$$\begin{aligned} &\sum_{j=0}^{2} \mathbb{E} \int_Q q_{j,\kappa}^{-1} |D^j y|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathbb{E} \int_{Q_0} o_\kappa^{-1} |u|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathbb{E} \int_Q r_\kappa^{-1} |U|^2 \,\mathrm{d}x\,\mathrm{d}t \\ &\qquad \le C_R \left(\varepsilon_\kappa^{-1} \mathbb{E} \|y_0\|_H^2 + \|\phi\|_{\mathcal{S}_\kappa}^2 \right). \end{aligned} \tag{4.6}$$

Here $D^0 y = y$, $D^1 y = \nabla y$ and $D^2 y$ is the full Hessian.

For fixed parameters, the difference form of (4.6) yields

$$\|D^j \delta y\|_{\mathcal{S}_\kappa}^2 \le C_R \rho_j(s, \kappa) \left(\varepsilon_\kappa^{-1} \mathbb{E} \|\delta y_0\|_H^2 + \|\delta \phi\|_{\mathcal{S}_\kappa}^2 \right), \qquad j = 0, 1, 2.$$

The unweighted estimate

$$\|(y, u, U)\|_{\mathcal{H} \times \mathcal{U}}^2 \le C \left[(1 + \varepsilon_\kappa^{-1}) \mathbb{E} \|y_0\|_H^2 + \|\phi\|_{\mathcal{S}_\kappa}^2 \right] \tag{4.7}$$

holds with C independent of s, κ and the data. It also holds for differences at fixed parameters.

Proof. Step 1: weight bounds. All constants are independent of $s \ge s_0$ and $\kappa \ge 1$, with λ fixed. On $(a, T) \times \mathcal{O}$,

$$\frac{F_s \sigma^{2j}}{B_s^*} = \lambda^{-2} \sigma^{-6+2j} \le c_j s^{-6+2j}, \qquad j = 0, 1, 2,$$

since ξ_* has a positive lower bound. The identity $\kappa^{-1+j/2} W_\kappa = \kappa^{-2+j/2} (\kappa W_\kappa)$ compares the early-time contribution to $q_{j,\kappa}$ with b_κ. This proves (4.3), including $t \le a$, where the Carleman terms vanish.

To prove the upper bounds, observe that

$$\frac{\alpha_*}{\xi_*} = 1 - e^{\lambda(2M - \eta)} \le -(e^{\lambda M} - 1) =: -d_\lambda < 0.$$

Thus, for every nonnegative power p in the weights,

$$(s\xi_*)^p F_s \le \sup_{r \ge 0} r^p e^{-2 d_\lambda r} < \infty.$$

It follows that $B_s^*, O_s^*, R_s^*, F_s \sigma^{2j}$ and m_s are uniformly bounded. For the early-time weights, use $\zeta(t) \le L_\zeta (c - t)$ for $t \le c$ and the boundedness of a linear factor times a decaying exponential. For $t \ge c$, $W_\kappa = 0$. Since $\kappa \ge 1$, the remaining powers of κ give bounded multiples of W_κ. Positivity follows from $W_\kappa > 0$ on $[0, b]$, $F_s > 0$ on (a, T), and $a < b$. Compactness of $J_t \times \overline{\mathcal{O}}$ gives $m_s \ge c_J s^6 e^{-C_J s}$, hence (4.4).

Step 2: observability at $z(0)$. Let (z, Z) solve the weak-source backward equation from Lemma 3.4. Since $W_\kappa(0) = \varepsilon_\kappa$, $W_\kappa(T) = 0$, and

$$W_\kappa' = \kappa W_\kappa + \frac{m_s}{4(1 + L_\zeta)} e^{-\kappa(c-t)} \zeta',$$

the variational Itô formula for $W_\kappa(t)\|z(t)\|_H^2$ gives

$$\varepsilon_\kappa\mathbb{E}\|z(0)\|_H^2+\kappa\mathbb{E}\int_Q W_\kappa|z|^2\,\mathrm{d}x\,\mathrm{d}t+2\mathbb{E}\int_Q W_\kappa|\Delta z|^2\,\mathrm{d}x\,\mathrm{d}t+\mathbb{E}\int_Q W_\kappa|Z|^2\,\mathrm{d}x\,\mathrm{d}t$$
$$=-\mathbb{E}\int_Q (W_\kappa'-\kappa W_\kappa)|z|^2\,\mathrm{d}x\,\mathrm{d}t-2\mathbb{E}\int_0^T W_\kappa\langle\mathfrak{D}\Phi,z\rangle_{V',V}\,\mathrm{d}t.$$

The V'–V pairing is integrable by Lemma 3.4. Localization and Lemma 2.3 justify taking zero expectation of the martingale term. Set $h_z=\|z\|_H$, $d_z=\|\Delta z\|_H$, and $p_j=\|\Phi_j\|_{L^2(\mathcal{O})}$, using component norms. For the pairings in $\mathfrak{D}\Phi$, $\|\nabla z\|_H^2\le h_z d_z$ and $\|D^2 z\|_H\le C_{\mathcal{O}} d_z$ give, by Young's inequality, uniformly in $\kappa\ge 1$,

$$2p_0h_z\le\frac{\kappa}{4}h_z^2+4\kappa^{-1}p_0^2,\qquad 2p_1\|\nabla z\|_H\le\frac{\sqrt{\kappa}}{4}\|\nabla z\|_H^2+4\kappa^{-1/2}p_1^2\le\frac{\kappa}{16}h_z^2+\frac{1}{4}d_z^2+4\kappa^{-1/2}p_1^2,$$
$$2p_2\|D^2z\|_H\le\frac{1}{2}d_z^2+2C_{\mathcal{O}}^2p_2^2.$$

The sum of these three bounds is at most $\frac{\kappa}{2}h_z^2+d_z^2+C(\kappa^{-1}p_0^2+\kappa^{-1/2}p_1^2+p_2^2)$. The cutoff term satisfies

$$-\frac{m_s}{4(1+L_\zeta)}e^{-\kappa(c-t)}\zeta'(t)\le\frac{m_s}{4}\mathbf{1}_{J_t}(t)\le\frac{1}{4}B_s^*(t,x)\mathbf{1}_{J_t}(t).$$

Consequently,

$$\varepsilon_\kappa\mathbb{E}\|z(0)\|_H^2+\frac{\kappa}{2}\mathbb{E}\int_Q W_\kappa|z|^2\,\mathrm{d}x\,\mathrm{d}t+\mathbb{E}\int_Q W_\kappa|\Delta z|^2\,\mathrm{d}x\,\mathrm{d}t+\mathbb{E}\int_Q W_\kappa|Z|^2\,\mathrm{d}x\,\mathrm{d}t$$
$$\le\frac{1}{4}\mathbb{E}\int_{J_t\times\mathcal{O}}B_s^*|z|^2\,\mathrm{d}x\,\mathrm{d}t+C\mathbb{E}\int_Q W_\kappa\big(\kappa^{-1}|\Phi_0|^2+\kappa^{-1/2}|\Phi_1|^2+|\Phi_2|^2\big)\,\mathrm{d}x\,\mathrm{d}t.\tag{4.8}$$

Apply Theorem 3.8 to (z,Z) on (a,T). The cutoff term in (4.8) is absorbed by one quarter of the late-time state integral. Adding the Carleman estimate to twice the early-time estimate gives the weight $B_s^*+\kappa W_\kappa$ and the initial term $2\varepsilon_\kappa\mathbb{E}\|z(0)\|_H^2$, with source weights bounded by constant multiples of $F_s\sigma^{2j}+\kappa^{-1+j/2}W_\kappa=q_{j,\kappa}$. Using $o_\kappa\ge O_s^*$ and $r_\kappa\ge R_s^*$ and discarding nonnegative terms yields (4.5), uniformly in both parameters.

Step 3: the Riesz construction. Fix s,κ and omit the weight subscripts. We adapt the Riesz construction of Lemma 3.6 to include the initial trace and three weak-source components. For $z_T\in L^2(\Omega,\mathcal{F}_T;H)$ and progressive unweighted L^2 sources, let $\mathfrak{E}$ be the closure of the vectors $e=(z|_{Q_0},\Phi_0,\Phi_1,\Phi_2,Z)$ generated by (3.10) in the progressive L^2 product with weights (o,q_0,q_1,q_2,r). The weight and energy bounds place these vectors in the product space. By (4.5), $e\mapsto(z(0),z)$ extends continuously to $\mathfrak{E}$ with values in $L^2(\Omega,\mathcal{F}_0;H)\times L^2_{\mathcal{P}}(\Omega\times Q;b)$, although closure elements need not be backward solutions.

Set $\mathcal{N}=\varepsilon^{-1}\mathbb{E}\|y_0\|_H^2+\|\phi\|_{\mathcal{S}_\kappa}^2$ and define on the generating vectors

$$\ell(e)=\mathbb{E}(y_0,z(0))_H+\mathbb{E}\int_Q\phi z\,\mathrm{d}x\,\mathrm{d}t,\qquad|\ell(e)|^2\le C_R\mathcal{N}\|e\|_{\mathfrak{E}}^2.$$

By weighted Cauchy–Schwarz, ℓ extends continuously to $\mathfrak{E}$. Let $\widehat{e} = (\widehat{\zeta}, \widehat{\Phi}_0, \widehat{\Phi}_1, \widehat{\Phi}_2, \widehat{Z})$ be its unique weighted Riesz representer and set

$$Y_0 = -q_0\widehat{\Phi}_0, \qquad Y_1 = q_1\widehat{\Phi}_1, \qquad Y_2 = -q_2\widehat{\Phi}_2, \qquad u = -o\widehat{\zeta}, \qquad U = -r\widehat{Z}.$$

The signs follow from (3.9). For the reciprocal-weight cost, we have $\|\widehat{e}\|_{\mathfrak{E}}^2 \le C_R\mathcal{N}$. We have obtained progressive representatives by closure and deterministic weighting; uniform weight bounds ensure unweighted square integrability. *Step 4: identification of the state.* Step 1 gives $\|\phi\|^2_{L^2_{\mathbb{F}}(0,T;H)} \le \|b\|_\infty \|\phi\|^2_{\mathcal{S}_\kappa}$. By Proposition 2.2, there is a state $y \in \mathcal{H}$ with initial datum y_0 and controls (u, U). Combining the Riesz identity with the duality formula (3.12), we obtain

$$\mathbb{E}(y(T), z_T)_H = \mathbb{E}\int_Q \big[(y - Y_0)\Phi_0 - (\nabla y - Y_1)\cdot\Phi_1 + (D^2 y - Y_2) : \Phi_2\big]\,\mathrm{d}x\,\mathrm{d}t.$$

Set $z_T = 0$ and take successively

$$(\Phi_0, \Phi_1, \Phi_2) = (y - Y_0, 0, 0), \quad (0, \nabla y - Y_1, 0), \quad (0, 0, D^2 y - Y_2).$$

These progressive L^2 sources are admissible by Step 3 and the derivative bounds for y, since $y \in L^2_{\mathbb{F}}(0,T;V)$; hence $y - Y_0$, $\nabla y - Y_1$, and $D^2 y - Y_2$ vanish. Since $y(T) \in L^2(\Omega, \mathcal{F}_T; H)$, testing with zero sources and $z_T = y(T)$ gives $\mathbb{E}\|y(T)\|_H^2 = 0$. Replacing Y_j by $D^j y$ in the Riesz bound proves (4.6).

Step 5: linearity and cost. The space $\mathfrak{E}$, its inner product and the identification maps are independent of the data. Linearity of ℓ in (y_0, ϕ) implies linearity of $\mathcal{R}_\kappa$. The difference form of (4.6) and $b^{-1} = (q_j/b)q_j^{-1}$ give the derivative bounds. The linear forward energy estimate, Step 3 and the uniform weight bounds give (4.7), uniformly in s, κ. We have thus constructed a bounded operator into $\mathcal{H} \times \mathcal{U}$ for fixed parameters. □

5. Null controllability with a full-domain diffusion control

Using the linear operator, we will select the drift source by contraction, independently of g, and recover the diffusion control by compensation.

Lemma 5.1. *Let $A \in \{A_N, A_C\}$ and assume only the conditions on F in Assumption 1.1(i). There exists a globally Lipschitz map*

$$\mathcal{K} : L^2(\Omega, \mathcal{F}_0; H) \longrightarrow \mathcal{H} \times \mathcal{U}, \qquad \mathcal{K}(0) = 0,$$

such that $\mathcal{K}(y_0) = (y, u, U)$ satisfies the variational system

$$\begin{cases} \mathrm{d}y + Ay\,\mathrm{d}t = \big(F[y] + Bu\big)\,\mathrm{d}t + U\,\mathrm{d}W_t, \\ \qquad y(0) = y_0, \qquad y(T) = 0. \end{cases} \tag{5.1}$$

The endpoint identities hold in $L^2(\Omega; H)$, and U is the free total diffusion. For any two initial data,

$$\|\mathcal{K}(y_{0,1}) - \mathcal{K}(y_{0,2})\|^2_{\mathcal{H}\times\mathcal{U}} \le Ce^{C\Lambda}\mathbb{E}\|y_{0,1} - y_{0,2}\|_H^2. \tag{5.2}$$

Here $C > 0$ depends only on $\mathcal{O}, \mathcal{O}_0, T$ and the boundary realization.

Proof. Step 1: the drift estimate. Fix $s \geq s_0$ and $\kappa \geq 1$. For each initial datum and $\phi \in \mathcal{S}_\kappa$, Proposition 4.1 gives

$$(y_\phi, u_\phi, U_\phi) = \mathcal{R}_\kappa(y_0, \phi).$$

Assumption 1.1(i) and $V \hookrightarrow H^2(\mathcal{O})$ imply that $F[\cdot] : V \to H$ is Lipschitz outside one $\mathbb{P} \otimes \mathrm{d}t$-null set. Using $F[0] = 0$ and Lemma 2.1, we obtain progressive compositions. Weighted Cauchy–Schwarz yields, almost everywhere,

$$\frac{|F[y] - F[\widetilde{y}]|^2}{b_\kappa} \leq \left(\sum_{j=0}^{2} L_j^2 \frac{q_{j,\kappa}}{b_\kappa} \right) \left(\sum_{j=0}^{2} q_{j,\kappa}^{-1} |D^j(y - \widetilde{y})|^2 \right).$$

Integrating over $\Omega \times Q$ against $\mathrm{d}\mathbb{P}(\omega)\,\mathrm{d}x\,\mathrm{d}t$ and using (4.3) and (4.6) for differences gives

$$\|F[y_1] - F[y_2]\|_{\mathcal{S}_\kappa}^2 \leq \delta^2 \left(\varepsilon_\kappa^{-1} \mathbb{E}\|y_{0,1} - y_{0,2}\|_H^2 + \|\phi_1 - \phi_2\|_{\mathcal{S}_\kappa}^2 \right),$$

$$\delta^2 = C_R \sum_{j=0}^{2} L_j^2 \left(c_j s^{-6+2j} + \kappa^{-2+j/2} \right). \tag{5.3}$$

Comparison with zero data, using $F[0] = 0$, puts the drift in $\mathcal{S}_\kappa$.

Step 2: parameters and fixed point. Let $C_\dagger \geq 1$ be at least as large as C_R and the three numbers $C_R c_j$. Choose

$$\begin{aligned} s &= \max\{s_0, (24C_\dagger)^{1/6} L_0^{1/3}, (24C_\dagger)^{1/4} L_1^{1/2}, (24C_\dagger)^{1/2} L_2\}, \\ \kappa &= \max\{1, (24C_\dagger)^{1/2} L_0, (24C_\dagger)^{2/3} L_1^{4/3}, 24C_\dagger L_2^2\}. \end{aligned} \tag{5.4}$$

Each term in δ^2 is at most $1/24$ by (5.4); terms with $L_j = 0$ vanish. Hence $\delta^2 \leq 1/4$, and $s + \kappa \leq C(1 + L_0 + L_1^{4/3} + L_2^2) = C\Lambda$. The parameters depend only on the Lipschitz bounds and geometry; we fix them for all initial data.

For fixed y_0, define $\mathcal{T}_{y_0}\phi = F[y_\phi]$. By $F[0] = 0$ and $\mathcal{R}_\kappa(0,0) = 0$, (5.3) makes $\mathcal{T}_{y_0}$ a self-map of $\mathcal{S}_\kappa$ with Lipschitz constant $\delta \leq 1/2$ and unique fixed point $\phi_*(y_0)$. Define

$$\mathcal{K}(y_0) = \mathcal{R}_\kappa\big(y_0, \phi_*(y_0)\big).$$

The fixed-point identity $\phi_*(y_0) = F[y]$ holds in $\mathcal{S}_\kappa$ and L^2 since b_κ is bounded. Thus the selected state satisfies (5.1) and the linear endpoint identities. For $y_0 = 0$, the fixed point is 0, so $\mathcal{K}(0) = 0$. The coefficient g has not entered the construction.

Step 3: cost and stability. Put $h^2 = \varepsilon_\kappa^{-1}\mathbb{E}\|y_0\|_H^2$. Comparison with zero data in (5.3) gives

$$\|\phi_*\|_{\mathcal{S}_\kappa}^2 \leq \delta^2 \left(h^2 + \|\phi_*\|_{\mathcal{S}_\kappa}^2 \right), \qquad \|\phi_*\|_{\mathcal{S}_\kappa}^2 \leq \frac{\delta^2}{1 - \delta^2} h^2 \leq h^2.$$

Thus (4.6) bounds the weighted cost by $2C_R h^2$, and (4.7) for (y_0, ϕ_*) gives

$$\|\mathcal{K}(y_0)\|_{\mathcal{H}\times\mathcal{U}}^2 \leq C(1 + \varepsilon_\kappa^{-1})\mathbb{E}\|y_0\|_H^2 \leq Ce^{C\Lambda}\mathbb{E}\|y_0\|_H^2. \tag{5.5}$$

Here we have used (4.4) and $s + \kappa \leq C\Lambda$. Keeping the parameters and Riesz inner product fixed, we apply (5.3) to two fixed points and absorb the difference term to obtain

$$\|\phi_{*,1} - \phi_{*,2}\|_{\mathcal{S}_\kappa}^2 \leq \frac{\delta^2}{1 - \delta^2} \varepsilon_\kappa^{-1} \mathbb{E}\|y_{0,1} - y_{0,2}\|_H^2.$$

The difference form of (4.7) and $1+\varepsilon_\kappa^{-1}\le Ce^{C\Lambda}$ yield (5.2) for $\mathcal{K}$. □

Proof of Theorem 1.3. Step 1: diffusion compensation. Given y_0, choose $(y,u,U)=\mathcal{K}(y_0)$ from Lemma 5.1 and define

$$v=U-g[y],\qquad \Gamma(y_0)=(y,u,v).$$

Since $y\in\mathcal{H}$, Lemma 2.1 shows that $g[y]$ is progressive and square integrable. Thus $(u,v)\in\mathcal{U}$ and $g[y]+v=U$, so (5.1) shows that y solves (1.1) in the variational sense, with zero terminal value in $L^2(\Omega;H)$ and almost surely in H. Proposition 2.2 gives uniqueness for these controls.

Step 2: cost. Write $v=U-(g[y]-g_*)-g_*$. Applying the squared triangle inequality twice gives

$$\begin{aligned}\|v\|^2_{L^2_{\mathbb{F}}(0,T;H)}&\le 4\|U\|^2_{L^2_{\mathbb{F}}(0,T;H)}+4\|g[y]-g_*\|^2_{L^2_{\mathbb{F}}(0,T;H)}+2\|g_*\|^2_{L^2_{\mathbb{F}}(0,T;H)}\\&\le 4\|U\|^2_{L^2_{\mathbb{F}}(0,T;H)}+C(1+b_g)\|y\|^2_{\mathcal{H}}+2\|g_*\|^2_{L^2_{\mathbb{F}}(0,T;H)}.\end{aligned}$$

In the last estimate, we have used (1.4) at zero, $a_g\le 2$ and the embeddings of $\mathcal{H}$ into $L^2_{\mathbb{F}}(0,T;V)$ and $L^2_{\mathbb{F}}(0,T;H)$. Combining it with (5.5), we obtain (1.5), with additive term $2\|g_*\|^2_{L^2_{\mathbb{F}}(0,T;H)}$.

Step 3: Lipschitz selection. For the same $\mathcal{K}$, denote differences by δ. The additive amplitude cancels: $\delta v=\delta U-(g[y_1]-g[y_2])$. Therefore

$$\|\delta v\|^2_{L^2_{\mathbb{F}}(0,T;H)}\le 2\|\delta U\|^2_{L^2_{\mathbb{F}}(0,T;H)}+2a_g\|\delta y\|^2_{L^2_{\mathbb{F}}(0,T;V)}+2b_g\|\delta y\|^2_{L^2_{\mathbb{F}}(0,T;H)}.$$

Combining this estimate with (5.2) and $a_g\le 2$ proves (1.6). The cost and Lipschitz constants are independent of C_E, uniform for $0\le a_g<2$, and have the stated b_g dependence. Finally, $\mathcal{K}(0)=0$ and $g[0]=g_*$ give $\Gamma(0)=(0,0,-g_*)$. □

6. Null controllability by a drift control

We will retain the prescribed diffusion by multiplying deterministic states and controls by a scalar stochastic amplitude. Set

$$\mathcal{X}_d=C([0,T];H)\cap L^2(0,T;V),\qquad \mathcal{U}_d=L^2(0,T;H_0),$$

where $\|w\|^2_{\mathcal{X}_d}=\sup_{t\le T}\|w(t)\|^2_H+\int_0^T\|w(t)\|^2_V\,\mathrm{d}t$. The space $\mathcal{X}_d$ is closed in $C([0,T];H)\times L^2(0,T;V)$ because the limits agree in $L^2(0,T;H)$. Hence $\mathcal{X}_d$ and $\mathcal{U}_d$ are separable Banach spaces. Write

$$F[z](t,x)=F(t,x,z(t,x),\nabla z(t,x),D^2z(t,x)).$$

Lemma 6.1. *Let $A\in\{A_N,A_C\}$ and suppose that F is deterministic and jointly measurable, vanishes at $(r,p,X)=0$, and satisfies (1.3). Then there exists a globally Lipschitz map $\Gamma_d:H\to\mathcal{X}_d\times\mathcal{U}_d$, $\Gamma_d(\zeta)=(w_\zeta,h_\zeta)$, with $\Gamma_d(0)=0$, such that*

$$\begin{cases}\partial_t w_\zeta+Aw_\zeta=F[w_\zeta]+Bh_\zeta,\\ \qquad w_\zeta(0)=\zeta,\qquad w_\zeta(T)=0.\end{cases}\tag{6.1}$$

The equation holds in V', with endpoints in H. For the selected control h_ζ, the state w_ζ is the unique variational solution, and

$$\|w_\zeta - w_{\widetilde{\zeta}}\|^2_{\mathcal{X}_d} + \|h_\zeta - h_{\widetilde{\zeta}}\|^2_{\mathcal{U}_d} \le Ce^{C\Lambda}\|\zeta - \widetilde{\zeta}\|^2_H, \tag{6.2}$$

where C depends only on $\mathcal{O}, \mathcal{O}_0, T$ and the boundary realization.

Proof. Step 1: linear selection. Fix s, κ and abbreviate the weights in Proposition 4.1 by b, o, q_j, ε. Define $\mathcal{S}_d = L^2(Q, b^{-1}\,\mathrm{d}t\,\mathrm{d}x)$, with norm $\|\phi\|^2_{\mathcal{S}_d} = \int_0^T \int_{\mathcal{O}} b(t,x)^{-1}|\phi(t,x)|^2\,\mathrm{d}x\,\mathrm{d}t$. For deterministic data $(\zeta, \phi) \in H \times \mathcal{S}_d$, let (Y, h^{st}, U) be the stochastic linear selection. Taking Bochner expectations, set

$$w = \mathbb{E}Y, \qquad h = \mathbb{E}h^{\mathrm{st}}, \qquad \mathcal{L}_d(\zeta, \phi) = (w, h).$$

The drift is integrable in V' and the H-valued stochastic integral has zero expectation, so

$$\partial_t w + Aw = \phi + Bh, \qquad w(0) = \zeta, \qquad w(T) = 0.$$

By the energy bound, Y is Bochner integrable in $\mathcal{X}_d$ and h^{st} in $\mathcal{U}_d$; traces and derivatives therefore commute with expectation. Boundedness of b gives $\mathcal{S}_d \subset L^2(Q)$. By Jensen's inequality with the deterministic weights, (4.6) yields

$$\sum_{j=0}^{2} \int_0^T \int_{\mathcal{O}} q_j^{-1}|D^j w|^2\,\mathrm{d}x\,\mathrm{d}t + \int_0^T \int_{\mathcal{O}_0} o^{-1}|h|^2\,\mathrm{d}x\,\mathrm{d}t \le C_R\left(\varepsilon^{-1}\|\zeta\|^2_H + \|\phi\|^2_{\mathcal{S}_d}\right), \tag{6.3}$$

with C_R enlarged if necessary. Jensen's inequality also transfers (4.7). Thus $\mathcal{L}_d$ is bounded and linear, with both estimates valid for differences. We have used expectation only in this linear step.

Step 2: fixed point and stability. For $(w, h) = \mathcal{L}_d(\zeta, \phi)$, define $\mathcal{T}_\zeta \phi = F[w]$. Measurability, Lipschitz continuity of F and $F[0] = 0$ make the composition measurable. The drift estimate in Lemma 5.1 applies to (6.3) with the ratios (4.3), giving

$$\|\mathcal{T}_\zeta\phi - \mathcal{T}_{\widetilde{\zeta}}\widetilde{\phi}\|^2_{\mathcal{S}_d} \le \delta^2\left(\varepsilon^{-1}\|\zeta - \widetilde{\zeta}\|^2_H + \|\phi - \widetilde{\phi}\|^2_{\mathcal{S}_d}\right),$$

where $\delta^2 = C_R\sum_{j=0}^2 L_j^2(c_j s^{-6+2j} + \kappa^{-2+j/2})$. Choose s, κ by (5.4), enlarging the constant so that $\delta \le 1/2$, and fix them for all initial data. Comparison with $(\widetilde{\zeta}, \widetilde{\phi}) = (0, 0)$ gives a self-map of $\mathcal{S}_d$; for fixed ζ, it has a unique fixed point ϕ_ζ.

Set $\Gamma_d(\zeta) = \mathcal{L}_d(\zeta, \phi_\zeta)$. The fixed-point identity yields (6.1), including the endpoint identities. The stability argument in Lemma 5.1, with the linear energy bound and $\varepsilon^{-1} \le Ce^{C(s+\kappa)} \le Ce^{C\Lambda}$, gives (6.2). Thus Γ_d is Borel measurable, and fixed-point uniqueness gives $\Gamma_d(0) = 0$. For solutions with the same control and initial datum, (B.5) and Gronwall's inequality give uniqueness. □

Proof of Theorem 1.8. Step 1: well-posedness. Write $G[z](t) = G(\omega, t, z(t))$. By joint $\mathcal{P} \otimes \mathcal{B}(H)$-measurability, $G[z]$ is progressive whenever the H-valued process z is progressive. By the common-null-set bounds in Assumption 1.7, the proof of Proposition 2.2 applies with $g = G$, $a_g = 0$, $b_g = L_G^2$ and $g_* = 0$. This gives a unique variational solution in $\mathcal{H}$ for every $u \in L^2_{\mathbb{F}}(0, T; H_0)$, without diffusion control.

Step 2: measurable selection. Let $(w,h)=\Gamma_d(y_0)$, where Γ_d is given by Lemma 6.1. Since the selection is continuous and the target spaces are separable, (w,h) is $\mathcal{F}_0$-measurable in $\mathcal{X}_d\times\mathcal{U}_d$. Continuous evaluation on $C([0,T];H)$ makes w adapted. For the L^2 coordinates, $\mathcal{F}_0$-simple approximations in $L^2(\Omega;L^2(0,T;V))$ and $L^2(\Omega;\mathcal{U}_d)$, with measurable time representatives and a $\mathbb{P}\otimes\mathrm{d}t$-almost everywhere convergent subsequence, give $\mathcal{F}_0\otimes\mathcal{B}([0,T])$-measurable, hence progressive, versions of h and of w as a V-valued process. The latter agrees with the continuous H-valued version $\mathbb{P}\otimes\mathrm{d}t$-almost everywhere.

Step 3: the scalar amplitude. Define the scalar coefficient

$$\sigma_w(\omega,t,r)=r\,b(\omega,t,rw(t,\omega)),\qquad r\in\mathbb{R}.$$

It is $\mathcal{P}\otimes\mathcal{B}(\mathbb{R})$-measurable because w is progressive. If $w(t,\omega)\neq 0$, the form $G(z)=b(z)z$ gives, for all $r,s\in\mathbb{R}$,

$$|\sigma_w(\omega,t,r)-\sigma_w(\omega,t,s)|=\frac{\|G(\omega,t,rw(t))-G(\omega,t,sw(t))\|_H}{\|w(t)\|_H}\le L_G|r-s|.$$

If $w(t,\omega)=0$, this difference is at most $B_0|r-s|$. These bounds hold outside one $\mathbb{P}\otimes\mathrm{d}t$-null set for all r,s, and $|\sigma_w(\omega,t,r)|\le B_0|r|$. The scalar SDE

$$\mathrm{d}M_t=\sigma_w(\omega,t,M_t)\,\mathrm{d}W_t,\qquad M_0=1,$$

has a unique global strong solution in $L^2_{\mathbb{F}}(\Omega;C([0,T];\mathbb{R}))$. This follows from Picard iteration with Lipschitz constant $\max\{L_G,B_0\}$. $\beta(t)=b(\omega,t,M_tw(t))$ is progressive and bounded by B_0. The identity $\mathrm{d}M_t=\beta(t)M_t\,\mathrm{d}W_t$ and uniqueness for the scalar linear equation give

$$M_t=\exp\left(\int_0^t\beta(s)\,\mathrm{d}W_s-\frac{1}{2}\int_0^t\beta(s)^2\,\mathrm{d}s\right)>0.$$

M is a continuous martingale. Boundedness of β also gives Novikov's condition for

$$N_t=\exp\left(2\int_0^t\beta(s)\,\mathrm{d}W_s-2\int_0^t\beta(s)^2\,\mathrm{d}s\right).$$

Thus $\mathbb{E}(N_t\mid\mathcal{F}_0)=1$. Since $M_t^2=N_t\exp(\int_0^t\beta(s)^2\,\mathrm{d}s)$,

$$\mathbb{E}(M_t^2\mid\mathcal{F}_0)\le e^{B_0^2t},\qquad\mathbb{E}\left(\sup_{t\le T}M_t^2\mid\mathcal{F}_0\right)\le 4e^{B_0^2T}.\tag{6.4}$$

The first bound follows from the exponential representation. For $\mathbf{1}_EM$ with any $E\in\mathcal{F}_0$, Doob's inequality gives the second. β may depend on $\mathcal{F}_0$; β may also depend on w and M.

Set $y=Mw$, $u=Mh$. Both are progressive, with y continuous in H. Conditioning on the $\mathcal{F}_0$-measurable pair (w,h), we obtain from (6.4)

$$\|y\|_{\mathcal{H}}^2+\|u\|_{L^2_{\mathbb{F}}(0,T;H_0)}^2\le 4e^{B_0^2T}\mathbb{E}\left(\|w\|_{\mathcal{X}_d}^2+\|h\|_{\mathcal{U}_d}^2\right).\tag{6.5}$$

Only second moments of y_0 are needed.

Step 4: the transformed equation. It follows from the deterministic equation that $\partial_tw=-Aw+F[w]+Bh\in L^2(\Omega;L^2(0,T;V'))$. For $\psi\in V$, the process $(w,\psi)_H$ is adapted and absolutely continuous, and hence has zero covariation with M. Set

$$\tau_m=\inf\{t\in[0,T]:M_t\ge m\}\wedge T,\qquad m\ge 2,$$

with $\inf \varnothing = \infty$. These stopping times increase to T almost surely. Positivity and spatial constancy of M give $D^j(Mw) = MD^j w$, $j = 0, 1, 2$, while homogeneity gives $MF[w] = F[y]$ outside the exceptional set in Assumption 1.7. Moreover, M preserves V and $\beta(t)y(t) = G[y](t)$. The Itô product formula up to τ_m gives the stopped variational equation. By (6.5) and Lipschitz continuity of F, $J = -Ay + F[y] + Bu$ belongs to $L^2_{\mathbb{F}}(0, T; V')$, and $G[y] \in L^2_{\mathbb{F}}(0, T; H)$ since $\|G[y](s)\|_H \le B_0 \|y(s)\|_H$. To remove the stopping times, we use the bounds

$$\mathbb{E} \sup_{t \le T} \left\| \int_0^t \mathbf{1}_{\{s > \tau_m\}} J(s)\, \mathrm{d}s \right\|_{V'}^2 \le T\mathbb{E} \int_0^T \mathbf{1}_{\{s > \tau_m\}} \|J(s)\|_{V'}^2\, \mathrm{d}s \longrightarrow 0,$$

$$\mathbb{E} \sup_{t \le T} \left\| \int_0^t \mathbf{1}_{\{s > \tau_m\}} G[y](s)\, \mathrm{d}W_s \right\|_H^2 \le 4\mathbb{E} \int_0^T \mathbf{1}_{\{s > \tau_m\}} \|G[y](s)\|_H^2\, \mathrm{d}s \longrightarrow 0.$$

Moreover, $y(\cdot \wedge \tau_m) \to y$ in $L^2(\Omega; C([0, T]; H))$ by dominated convergence. A subsequence and a countable dense subset of V yield the identity for all $t \in [0, T]$ and $\psi \in V$ on one event of probability one. Thus y solves (1.7) with diffusion $G[y]$, and $y(0) = y_0$, $y(T) = M_T w(T) = 0$ almost surely in H and in $L^2(\Omega; H)$. Estimates (6.5) and (6.2) at $\widetilde{\zeta} = 0$ give (1.8), with a constant depending only on $\mathcal{O}, \mathcal{O}_0, T$ and the boundary realization. The cost is independent of L_G; uniqueness follows from Step 1. □

Proof of Corollary 1.11. For $b(\omega, t, z) = \beta_0(\omega, t)$, all initial data share the amplitude

$$M_t = \exp\left(\int_0^t \beta_0(s)\, \mathrm{d}W_s - \frac{1}{2} \int_0^t \beta_0(s)^2\, \mathrm{d}s \right).$$

Fix β_0, which may depend on $\mathcal{F}_0$. For $(w_i, h_i) = \Gamma_d(y_{0,i})$ and $(y_i, u_i) = (Mw_i, Mh_i)$, $y_1 - y_2 = M(w_1 - w_2)$ and $u_1 - u_2 = M(h_1 - h_2)$. Applying (6.5) to these differences and using (6.2) gives

$$\|y_1 - y_2\|_{\mathcal{H}}^2 + \|u_1 - u_2\|_{L^2_{\mathbb{F}}(0,T;H_0)}^2 \le C e^{B_0^2 T + C\Lambda} \mathbb{E} \|y_{0,1} - y_{0,2}\|_H^2. \tag{6.6}$$

The selection $\Gamma_{\mathrm{sc}}(y_0) = (y, u)$ vanishes at zero since $\Gamma_d(0) = 0$. Its global Lipschitz continuity follows from (6.6). For state-dependent b, the amplitude generally depends on the initial datum, so this argument with a common multiplier does not apply. □

Remark 6.2. The coefficient $b(\omega, t, z)$ may depend on the entire state z and is spatially constant for fixed (ω, t, z). Thus the construction allows nonlocal z-dependence; it does not cover general pointwise diffusion $G[z](x) = \sigma(z(x))$. The drift must be deterministic and positively homogeneous of degree one; otherwise, $M^{-1}F(t, x, Mw, M\nabla w, MD^2 w)$ is generally random, and solving the transformed terminal-value problem pathwise may yield controls depending on future noise.

For spatially varying noise, take smooth $\beta = \beta(x)$ and set

$$M(t, x) = \exp\left(\beta(x) W_t - \frac{1}{2} \beta(x)^2 t \right).$$

With $[\Delta^2, M]w = \Delta^2(Mw) - M\Delta^2 w$, the product rule gives

$$M^{-1}[\Delta^2, M]w = 4M^{-1}\nabla M \cdot \nabla \Delta w + 2M^{-1}(\Delta M)\Delta w \\ + 4M^{-1}D^2M : D^2 w + 4M^{-1}\nabla \Delta M \cdot \nabla w + M^{-1}(\Delta^2 M)w.$$

The transformed drift therefore contains a third-order random term. For $y = Mw$, the Navier conditions read as follows in w:

$$w = 0, \qquad \Delta w + 2M^{-1}(\partial_\nu M)(\partial_\nu w) = 0 \quad \text{on } \Sigma.$$

Under clamped conditions, $Mw = \partial_\nu(Mw) = 0$ is equivalent to $w = \partial_\nu w = 0$, since $M > 0$.

This reduction does not cover spatially varying noise.

7. Local null controllability for superlinear drifts

To prove Theorem 1.14, we will adapt the source-term construction of [21]. The resulting causal control operator provides strong trajectory estimates for the polynomial derivative terms. Put $Q = (0,T) \times \mathcal{O}$. Elliptic regularity gives

$$D(A_N) = \{z \in H^4(\mathcal{O}) : z = \Delta z = 0 \text{ on } \partial\mathcal{O}\}, \qquad D(A_C) = H^4(\mathcal{O}) \cap H_0^2(\mathcal{O}).$$

The graph norms of $V = D(A^{1/2})$ and $D(A)$ are equivalent to the H^2 and H^4 norms, respectively.

7.1. Linear null controllability and maximal regularity.

Lemma 7.1. *For each $0 < h \le T$, there exists a deterministic bounded linear map $K_h : H \to L^2(0,h;H_0)$ such that the solution of $z' + Az = BK_h x$, $z(0) = x$, satisfies $z(h) = 0$ and*

$$\|K_h x\|_{L^2(0,h;H_0)} \le C_0 e^{C_1/h} \|x\|_H . \tag{7.1}$$

The constants C_0, C_1 depend only on $\mathcal{O}, \mathcal{O}_0$, the boundary realization and the fixed upper bound T for h.

Proof. Denote the spectral projector of A on $(0,\Lambda]$ by Π_Λ. For either boundary realization,

$$\|w\|_H \le C e^{C\Lambda^{1/4}} \|B^* w\|_{H_0}, \qquad w \in \operatorname{Ran} \Pi_\Lambda.$$

For $A_N = (-\Delta_D)^2$, the biharmonic cutoff Λ corresponds to Laplace eigenvalues below $\Lambda^{1/2}$, so the Laplace spectral inequality gives this bound. For A_C, see [18, Theorem 1.3]. Since A is nonnegative and self-adjoint and $B : H_0 \to H$ is bounded, [26, Theorem 2.8], with exponent $1/4$, gives final-state observability. Duality then yields (7.1) for $0 < h \le T$ after enlarging the constants.

For fixed h, define the bounded endpoint map

$$\mathcal{W}_h q = \int_0^h e^{-(h-s)A} B q(s)\, \mathrm{d}s.$$

The closed affine subspace $\{q : \mathcal{W}_h q = -e^{-hA}x\}$ of $L^2(0,h;H_0)$ is nonempty by null controllability. Its unique least-norm element $K_h x$ is characterized by

$$\mathcal{W}_h K_h x = -e^{-hA}x, \qquad K_h x \perp \ker \mathcal{W}_h.$$

Thus K_h is linear, and (7.1) follows by minimality. No closed-range assumption on $\mathcal{W}_h$ is needed. □

Lemma 7.2. *For $x \in V$, $f \in L^2(0,h;H)$ and $h \le T$, the solution of $z' + Az = f$, $z(0) = x$, satisfies*

$$\|z\|_{C([0,h];V)} + \|z\|_{L^2(0,h;D(A))} + \|z'\|_{L^2(0,h;H)} \le M_T\big(\|x\|_V + \|f\|_{L^2(0,h;H)}\big), \tag{7.2}$$

where M_T depends only on $\mathcal{O}, T$ and the boundary realization, uniformly in h. The selected homogeneous null trajectory therefore satisfies

$$\|z\|_{C([0,h];V)} + \|z\|_{L^2(0,h;D(A))} + \|z'\|_{L^2(0,h;H)} + \|K_h x\|_{L^2(0,h;H_0)} \le D_T e^{C_1/h}\,\|x\|_V\,. \tag{7.3}$$

Here D_T may also depend on $\mathcal{O}_0, C_0, C_1$ and is independent of h.

Proof. Let Π_m project onto the first m eigenfunctions of A. Testing the equation with data $\Pi_m x, \Pi_m f$ against Az_m and applying Young's inequality, we obtain, for $0 \le t \le h$,

$$\|z_m(t)\|_V^2 + \int_0^t \|Az_m(s)\|_H^2\,\mathrm{d}s \le \|\Pi_m x\|_V^2 + \int_0^t \|\Pi_m f(s)\|_H^2\,\mathrm{d}s.$$

Since the first eigenvalue is positive, the graph norms are controlled by $\|A^{1/2} z_m\|_H$ and $\|Az_m\|_H$. The identity $z_m' = \Pi_m f - Az_m$ gives the derivative bound. The estimates for $z_m - z_\ell$, together with $\Pi_m x \to x$ in V and $\Pi_m f \to f$ in $L^2(0,h;H)$, give convergence in all three spaces. The limit is the mild solution and satisfies the equation in H almost everywhere. Thus (7.2) holds uniformly for $0 < h \le T$. For $x \in V$, take $f = BK_h x$ and use boundedness of B, (7.1) and $V \hookrightarrow H$ to obtain (7.3) with a larger D_T. □

7.2. **A causal construction for decaying sources.** Put $p_* := \min_{j\in J} p_j > 1$. Fix $\beta = 1$ and choose a geometric ratio $r > 1$ with $\theta := r^\beta < \sqrt{p_*}$. Set

$$t_k = T(1 - r^{-k}), \qquad I_k = (t_k, t_{k+1}), \qquad h_k = (1 - r^{-1})Tr^{-k}, \qquad d_k = (T - t_k)^{-\beta}.$$

Choose $P_y, P_f > 0$ such that

$$\theta^2 P_y + C_1(1 - r^{-1})^{-\beta}\theta < P_f < p_* P_y. \tag{7.4}$$

Choose first $P_y > C_1(1 - r^{-1})^{-\beta}\theta/(p_* - \theta^2)$. The required interval is then nonempty. The parameters r, P_y, P_f depend only on p_* and C_1, independently of the random coefficients and diffusion parameters. Define

$$\rho_y(t) = \exp\left(-\frac{P_y}{(T-t)^\beta}\right), \qquad \rho_f(t) = \exp\left(-\frac{P_f}{(T-t)^\beta}\right), \quad t < T,$$

and set both weights to zero at T. They are positive and decreasing before T and satisfy

$$\frac{\rho_y(t)^{p_j}}{\rho_f(t)} \le 1 \quad (j \in J), \qquad \rho_f(t) \le \rho_y(t) \le 1. \tag{7.5}$$

Calligraphic letters denote pathwise weighted spaces; bold letters denote their adapted counterparts. Define the deterministic spaces

$$\begin{aligned} \|\phi\|_{\mathcal{S}} &= \|\phi/\rho_f\|_{L^2(0,T;H)}\,, \qquad \|u\|_{\mathcal{V}} = \|u/\rho_y\|_{L^2(0,T;H_0)}\,, \\ \|y\|_{\mathcal{X}} &= \|y/\rho_y\|_{C([0,T];V)} + \|y/\rho_y\|_{L^2(0,T;D(A))} + \|y'/\rho_y\|_{L^2(0,T;H)}\,. \end{aligned} \tag{7.6}$$

In $\mathcal{X}$, we require y/ρ_y to extend continuously to T with value zero and $y \in H^1(0,T;H)$. The derivative term is that of y, not y/ρ_y. The source and control spaces are isometric to separable L^2 spaces. For a state-space Cauchy sequence y_m with component limits Y_0, Y_1, Z, $Y_0 = Y_1$ in $L^2(0,T;H)$. Set $y = \rho_y Y_0$. Since $\rho_y \le 1$, $y_m \to y$ and $y_m' \to \rho_y Z$ in $L^2(0,T;H)$, so $y \in H^1(0,T;H)$ and $y' = \rho_y Z$. The limit Y_0 vanishes at T. This proves completeness and separability without differentiating y/ρ_y.

Let $\mathbf{S}$ and $\mathbf{V}$ be the progressive subspaces of $L^\infty(\Omega;\mathcal{S})$ and $L^\infty(\Omega;\mathcal{V})$. Let $\mathbf{X}$ be the adapted subspace of $L^\infty(\Omega;\mathcal{X})$, with its continuous V-valued representatives. We use strongly measurable Bochner spaces. Norm convergence implies convergence in $L^2(\Omega\times(0,T))$, whose progressive subspace is closed. State convergence is uniform in V outside a null set. Hence the adapted subspaces are closed Banach spaces.

Proposition 7.3 (Adapted source control)**.** *There exists a bounded linear map*

$$\mathcal{R} = (\mathcal{R}^y, \mathcal{R}^u) : L^\infty(\Omega, \mathcal{F}_0; V) \times \mathbf{S} \longrightarrow \mathbf{X} \times \mathbf{V}$$

whose output satisfies, pathwise,

$$y' + Ay = \phi + Bu, \qquad y(0) = x, \qquad y(T) = 0. \tag{7.7}$$

For some $M \ge 1$ depending only on $\mathcal{O}, \mathcal{O}_0, T$, the boundary realization and the fixed weight parameters,

$$\|y\|_{\mathbf{X}} + \|u\|_{\mathbf{V}} \le M\big(\|x\|_{L^\infty(\Omega;V)} + \|\phi\|_{\mathbf{S}}\big). \tag{7.8}$$

The same bound holds for almost every path in the deterministic norms of $V, \mathcal{S}, \mathcal{X}$ and $\mathcal{V}$. For the chosen weights, the additional dependence is only on p_*; M is independent of the source, initial datum, K_j and diffusion coefficient. The construction is causal: equal initial data and sources up to t give equal states and controls up to that time.

Proof. Step 1: Interval construction. Fix $x \in V$ and $\phi \in \mathcal{S}$. On I_k, we solve the uncontrolled equation with zero initial value:

$$w_k(t) = \int_{t_k}^{t} e^{-(t-s)A}\phi(s)\,\mathrm{d}s, \qquad c_k = w_k(t_{k+1}).$$

Lemma 7.2 gives continuity of w_k in V and $c_k \in V$. On the next interval, I_{k+1}, the translated control $K_{h_{k+1}}c_k$ cancels this terminal value. Define z_k by

$$z_k' + Az_k = B(K_{h_{k+1}}c_k)(\,\cdot - t_{k+1}), \qquad z_k(t_{k+1}) = c_k, \qquad z_k(t_{k+2}) = 0.$$

Let z_{-1} be the null trajectory on I_0 with initial value x. Define

$$y = \begin{cases} z_{-1} + w_0 & \text{on } I_0, \\ z_{k-1} + w_k & \text{on } I_k,\ k \ge 1, \end{cases} \qquad u(t) = \begin{cases} (K_{h_0}x)(t) & t \in I_0, \\ (K_{h_k}c_{k-1})(t - t_k) & t \in I_k,\ k \ge 1. \end{cases} \tag{7.9}$$

At t_k, $k \geq 1$, both traces of y equal c_{k-1}: the previous controlled trajectory ends at zero and $w_k(t_k) = 0$. Thus y is continuous in V on $[0, T)$, starts at x and satisfies (7.7) on compact subintervals of $[0, T)$. The construction is linear in (x, ϕ).

Step 2: Weighted estimates. Set $F_k = \|\phi/\rho_f\|_{L^2(I_k;H)}$. Then

$$\sum_{k\geq 0} F_k^2 = \int_0^T \left\| \frac{\phi(t)}{\rho_f(t)} \right\|_H^2 \mathrm{d}t = \|\phi\|_{\mathcal{S}}^2, \qquad \|\phi\|_{L^2(I_k;H)} \leq \rho_f(t_k) F_k. \tag{7.10}$$

For an interval I, let $|z|_{\mathcal{X}(I)}$ denote (7.6) restricted to I, including traces. Since ρ_y is decreasing, weighted norms are bounded by unweighted norms divided by ρ_y at the right endpoint. By Lemma 7.2 and (7.10),

$$|w_k|_{\mathcal{X}(I_k)} \leq M_T \frac{\rho_f(t_k)}{\rho_y(t_{k+1})} F_k = M_T e^{-(P_f - \theta P_y) d_k} F_k. \tag{7.11}$$

Similarly, (7.3) and the endpoint bound for c_k give

$$\begin{aligned} |z_k|_{\mathcal{X}(I_{k+1})} + \left\| (K_{h_{k+1}} c_k)/\rho_y \right\|_{L^2(I_{k+1};H_0)} &\leq D_T M_T \frac{e^{C_1 h_{k+1}^{-\beta}} \rho_f(t_k)}{\rho_y(t_{k+2})} F_k \\ &= D_T M_T e^{-[P_f - \theta^2 P_y - C_1 (1 - r^{-1})^{-\beta} \theta] d_k} F_k. \end{aligned} \tag{7.12}$$

The control is understood after time translation; since $\beta = 1$, the exponential cost is that in (7.3). Set

$$\eta_0 = P_f - \theta P_y > 0, \qquad \eta_1 = P_f - \theta^2 P_y - C_1 (1 - r^{-1})^{-\beta} \theta > 0.$$

Both gaps are positive by (7.4). The denominator is evaluated at the control endpoint t_{k+2}, rather than t_{k+1}: $h_{k+1}^{-\beta} = (1 - r^{-1})^{-\beta} \theta d_k$ and $(T - t_{k+2})^{-\beta} = \theta^2 d_k$. The initial component obeys the same estimate.

Each interval contains at most two components. Summing their squared L^2 norms and interval suprema, using $(a + b)^2 \leq 2a^2 + 2b^2$ and (7.10), we obtain

$$\|y\|_{\mathcal{X}} + \|u\|_{\mathcal{V}} \leq C_T \left[e^{(\theta P_y + C_1 (1 - r^{-1})^{-\beta}) T^{-\beta}} \|x\|_V + \|\phi\|_{\mathcal{S}} \right].$$

The separate bounds involve fixed multiples of $\|\phi\|_{\mathcal{S}}$ and $\|x\|_V$. Taking essential suprema gives the Bochner estimate with a larger M.

Step 3: Terminal value and global equation. Since (F_k) is bounded, $d_k \to \infty$, and η_0, η_1 are positive, the component estimates give

$$\sup_{t \in \overline{I}_k} \|y(t)/\rho_y(t)\|_V \longrightarrow 0.$$

Continuity of y and $\rho_y(t_k) > 0$ give continuity of the quotient at each junction. It extends continuously to T with value zero, so $y(T) = 0$.

The unweighted L^2 norms of Ay and the piecewise derivative are bounded because $\rho_y \leq 1$. The traces match on $(0, T)$, so $y \in H^1(0, T; H)$ and $y' = \phi + Bu - Ay$ almost everywhere. Integrability extends the integral identity from times before T to T. We have thus constructed a state in $\mathcal{X}$.

Step 4: Adaptedness and causality. For stochastic data, apply the construction on a common probability-one set. For each $t \in [t_k, t_{k+1}]$, $\phi|_{(t_k,t)}$ is $\mathcal{F}_t$-measurable in $L^2(t_k, t; H)$. By Lemma 7.2, $w_k(t)$ is $\mathcal{F}_t$-measurable in V and c_k is $\mathcal{F}_{t_{k+1}}$-measurable.

The bounded map $K_{h_{k+1}}$ sends c_k to an $\mathcal{F}_{t_{k+1}}$-measurable $L^2(0, h_{k+1}; H_0)$ random variable. By simple approximation, we obtain a jointly $\mathcal{F}_{t_{k+1}} \otimes \mathcal{B}(I_{k+1})$-measurable translated representative. The control is therefore known at t_{k+1} and is progressive on I_{k+1}; the trajectory z_k is adapted and continuous in V. On I_0, x is $\mathcal{F}_0$-measurable. Countable concatenation gives a progressive control and a continuous adapted state. Bounded linearity preserves strong Bochner measurability, so the output belongs to $\mathbf{X} \times \mathbf{V}$ and satisfies (7.8).

The control on I_k uses only x and the source before t_k, whereas $w_k(t)$ uses the source on (t_k, t). Thus equal data up to t yield equal states up to t and equal controls almost everywhere before t. □

7.3. **Polynomial nonlinearities.**

Lemma 7.4 (Space–time interpolation)**.** *Let $j \in \{0, 1, 2\}$ and $2 < q < \infty$. Assume $q \le q_j := 2(n+4)/(n-4+2j)$ when $n - 4 + 2j > 0$; when $n - 4 + 2j \le 0$, there is no additional restriction on q. Then every $z \in L^\infty(0, T; V) \cap L^2(0, T; D(A))$ satisfies*

$$\int_0^T \int_{\mathcal{O}} |D^j z(t, x)|^q \, \mathrm{d}x \, \mathrm{d}t \le C_q \, \|z\|^{q-2}_{L^\infty(0,T;V)} \int_0^T \|z(t)\|^2_{D(A)} \, \mathrm{d}t. \tag{7.13}$$

The finite endpoint $q = q_j$ is included. The constant depends only on $\mathcal{O}, j, q$ and the boundary realization, not on T or z.

Proof. Set $\vartheta = 2/q \in (0, 1)$ and $s = 2 + 2\vartheta$. Interpolation between $H^2(\mathcal{O})$ and $H^4(\mathcal{O})$, the assumed range of q and Sobolev embedding, including the finite endpoint, give

$$\left\| D^j z(t) \right\|_{L^q(\mathcal{O})} \le C_q \, \|z(t)\|_V^{1-\vartheta} \, \|z(t)\|^{\vartheta}_{D(A)} .$$

Raise this estimate to the q-th power and integrate. Since $q\vartheta = 2$ and $q(1-\vartheta) = q - 2$, we obtain (7.13). Only $L^2(0, T; H^4(\mathcal{O}))$ regularity is used; no pointwise Hessian bound is needed. □

Lemma 7.5 (Weighted nonlinear estimate)**.** *Under (1.9) and (7.5), we have $\mathcal{N} : \mathbf{X} \to \mathbf{S}$, $\mathcal{N}(0) = 0$, and*

$$\|\mathcal{N}(y) - \mathcal{N}(z)\|_{\mathbf{S}} \le C \sum_{j \in J} K_j \big(\|y\|_{\mathbf{X}}^{p_j - 1} + \|z\|_{\mathbf{X}}^{p_j - 1} \big) \, \|y - z\|_{\mathbf{X}} . \tag{7.14}$$

On a ball of radius R, the Lipschitz constant is at most $C \sum_{j \in J} K_j R^{p_j - 1}$ and tends to zero as $R \downarrow 0$. Here C depends only on $\mathcal{O}$, the boundary realization and the finitely many exponents p_j.

Proof. Fubini's theorem gives a common probability-one set on which the increment bounds hold for a.e. (t, x) and all arguments. On each such path, put $Y = y/\rho_y$ and $Z = z/\rho_y$

in $L^\infty(0,T;V)\cap L^2(0,T;D(A))$. Since the weights are independent of the spatial variable, $D^j y=\rho_y D^j Y$ and

$$\frac{|f_j(D^j y)-f_j(D^j z)|}{\rho_f}\le K_j\frac{\rho_y^{p_j}}{\rho_f}(|D^jY|^{p_j-1}+|D^jZ|^{p_j-1})|D^j(Y-Z)|.$$

By (7.5) and Hölder's inequality with exponents $2p_j/(p_j-1)$ and $2p_j$, we obtain

$$\left\|\frac{f_j(D^j y)-f_j(D^j z)}{\rho_f}\right\|_{L^2(Q)}\le K_j\left(\left\|D^jY\right\|_{L^{2p_j}(Q)}^{p_j-1}+\left\|D^jZ\right\|_{L^{2p_j}(Q)}^{p_j-1}\right)\left\|D^j(Y-Z)\right\|_{L^{2p_j}(Q)}.$$

Since $p_j\le p_j^*$, Lemma 7.4 applies with $q=2p_j$, including the finite endpoint, and gives

$$\|D^jY\|_{L^{2p_j}(Q)}\le C\|Y\|_{L^\infty(0,T;V)}^{1-1/p_j}\|Y\|_{L^2(0,T;D(A))}^{1/p_j}\le C\|y\|_{\mathcal{X}}.$$

Apply the same bound to Z and $Y-Z$, sum over j, and take the essential supremum in ω. This yields (7.14). With $z=0$ and $f_j(0)=0$, this also gives the source estimate.

Continuous adapted V-valued states are progressive. Simple approximations using $D^j: V\to L^2(\mathcal{O};E_j)$ give jointly progressive spatial representatives. Measurability of f_j and continuity in its last argument preserve measurability under composition. The estimate gives an $L^2(0,T;H)$ representative after division by ρ_f; separability of $L^2(0,T;H)$ gives strong Bochner measurability. Hence $\mathcal{N}(y)\in\mathbf{S}$.

□

7.4. **The nonlinear control construction.**

Proof of Theorem 1.14: existence and bounds. Let $M\ge 1$ be as in Proposition 7.3. Since $p_*\le p_j$ and $R^{p_j-1}\le R^{p_*-1}$ for $R\le 1$, Lemma 7.5 gives a deterministic $C_*>0$ such that

$$\|\mathcal{N}(y)\|_{\mathbf{S}}\le C_*R^{p_*-1}\|y\|_{\mathbf{X}},\qquad \|\mathcal{N}(y)-\mathcal{N}(z)\|_{\mathbf{S}}\le 2C_*R^{p_*-1}\|y-z\|_{\mathbf{X}}$$

whenever $0<R\le 1$ and both state norms are at most R. The same inequalities hold pathwise in the deterministic norms. Choose

$$R=\min\{1,(8MC_*)^{-1/(p_*-1)}\},\qquad \delta_T=R/(2M).$$

For $d=\|y_0\|_{L^\infty(\Omega;V)}\le\delta_T$, define $\Phi(y)=\mathcal{R}^y(y_0,\mathcal{N}(y))$ on the closed ball of radius R in the Banach space $\mathbf{X}$. Combining the preceding estimates with (7.8), we obtain

$$\|\Phi(y)\|_{\mathbf{X}}\le Md+MC_*R^{p_*}\le 5R/8,\qquad \|\Phi(y)-\Phi(z)\|_{\mathbf{X}}\le\tfrac14\|y-z\|_{\mathbf{X}}.$$

The contraction principle gives a unique fixed point of Φ in this ball. The iterates from zero are continuous and adapted, as is their $\mathbf{X}$ limit. Set $u=\mathcal{R}^u(y_0,\mathcal{N}(y))$. Pathwise,

$$y'+Ay=\mathcal{N}(y)+Bu,\qquad y(0)=y_0,\qquad y(T)=0.$$

On the radius-R ball, (7.8) and $MC_*R^{p_*-1}\le 1/8$ allow us to absorb the nonlinear term and obtain

$$\|y\|_{\mathbf{X}}+\|u\|_{\mathbf{V}}\le\tfrac87 Md\le 2Md.$$

The weights are at most one, so this also bounds the unweighted state, derivative and control norms. Hence $y\in L^\infty(\Omega;\mathfrak{X}_T)$ and (1.10) holds.

Define the diffusion control by $v = -g[y]$. The measurability in Assumption 1.1(ii) makes v progressive. Moreover,

$$\|g[y]\|_H^2 \le 2a_g \|y\|_V^2 + 2b_g \|y\|_H^2 + 2\|g_*\|_H^2 .$$

Integration in time and probability, using $a_g \le 2$ and (1.10), gives (1.11). In particular, $(u, v) \in \mathcal{U}$. Since $g[y] + v$ vanishes, the pathwise H-valued equation is (1.1), with terminal value zero. The choices of R, δ_T and the bounds for (y, u) are independent of diffusion parameters; $1 + b_g$ accounts for the cost of v.

For two admissible initial states, both trajectories lie in the radius-R ball. Linearity of $\mathcal{R}$, the nonlinear difference estimate and absorption give Lipschitz dependence of (y, u) in the weighted and hence unweighted norms. For the diffusion controls, (1.4) gives the corresponding Lipschitz dependence.

We have proved uniqueness of the selection in the weighted ball. For prescribed controls, Proposition 7.7 gives uniqueness in the stated strong-solution class. □

Lemma 7.6 (Energy estimate for coefficient terms)**.** *Let $j \in \{0, 1, 2\}$, $r_j = (n + 4)/(4 - j)$, and $a \in L^{r_j}(\mathcal{O})$, $a \ge 0$. For every $e \in V$ and $\varepsilon > 0$,*

$$\int_{\mathcal{O}} a(x)|D^j e(x)||e(x)|\,\mathrm{d}x \le \varepsilon \|e\|_V^2 + C_\varepsilon \|a\|_{L^{r_j}(\mathcal{O})}^{r_j} \|e\|_H^2 . \tag{7.15}$$

Proof. Hölder and Gagliardo–Nirenberg inequalities give

$$\int_{\mathcal{O}} a(x)|D^j e(x)||e(x)|\,\mathrm{d}x \le C \|a\|_{L^{r_j}(\mathcal{O})} \|e\|_V^{\theta_j} \|e\|_H^{2-\theta_j}, \qquad \theta_j = \frac{2(n + j)}{n + 4} < 2.$$

The H^2 norms on the form domain are controlled by $\|e\|_V$; Poincaré's inequality absorbs lower-order terms.

Apply Young's inequality with exponents $2/\theta_j$ and $2/(2 - \theta_j)$. The latter equals $r_j = (n + 4)/(4 - j)$, and (7.15) follows. The constant depends only on ε, the domain, j and the boundary realization. □

Proposition 7.7. *Suppose that Assumptions 1.13 and 1.1(ii) hold. For prescribed controls $(u, v) \in \mathcal{U}$ and initial datum $y_0 \in L^\infty(\Omega, \mathcal{F}_0; V)$, (1.1) has at most one strong solution in H whose paths belong almost surely to $C([0, T]; V) \cap L^2(0, T; D(A))$.*

Proof. Let y, z be solutions with the same initial state, controls, and Brownian motion, and put $e = y - z$. We show that the solutions are indistinguishable.

Step 1: Coefficient integrability. The progressive V-valued spectral projections converge in V at every time on the common probability-one set of V-continuous paths. Completeness of the filtration permits extension by zero outside this set. Both solutions therefore have progressive versions, and the maps $D^j : V \to L^2(\mathcal{O}; E_j)$ give jointly progressive spatial representatives. The increment condition yields, almost everywhere,

$$|\mathcal{N}(y) - \mathcal{N}(z)| \le \sum_{j \in J} a_j |D^j e|, \qquad a_j = K_j(|D^j y|^{p_j - 1} + |D^j z|^{p_j - 1}).$$

For almost every path, $a_j \in L^{r_j}(Q)$ with $r_j = (n+4)/(4-j)$. Indeed, for $p_j^* < \infty$,

$$(p_j - 1)r_j \le (p_j^* - 1)r_j = 2p_j^*.$$

If $(p_j - 1)r_j > 2$, apply Lemma 7.4, including at the endpoint; otherwise, $D^j y, D^j z \in L^2(Q)$ and finite $|Q|$ give the required integrability. For $p_j^* = \infty$, choose a finite admissible exponent above $\max\{2, (p_j - 1)r_j\}$. Interpolation at $2p_j$ also gives $\mathcal{N}(y), \mathcal{N}(z) \in L^2(0,T;H)$ pathwise.

Step 2: Localized energy estimate. Subtracting the equations cancels the prescribed controls:

$$\mathrm{d}e + Ae\,\mathrm{d}t = (\mathcal{N}(y) - \mathcal{N}(z))\,\mathrm{d}t + (g[y] - g[z])\,\mathrm{d}W_t, \qquad e(0) = 0.$$

The drift is locally H-integrable by Step 1 and $L^2(0,T;D(A))$ regularity, while (1.4) gives a locally square-integrable diffusion. Applying the localized Hilbert space Itô formula with $(Ae, e)_H = \|e\|_V^2$ and the form estimate with $\varepsilon = (2 - a_g)/(4|J|)$, we obtain

$$\mathrm{d}\,\|e\|_H^2 + c_0\,\|e\|_V^2\,\mathrm{d}t \le c(t)\,\|e\|_H^2\,\mathrm{d}t + 2\big(e, g[y] - g[z]\big)_H\,\mathrm{d}W_t,$$

where $c_0 = (2 - a_g)/2$ and one may take

$$c(t) = b_g + 2\sum_{j\in J} C_\varepsilon\,\|a_j(t)\|_{L^{r_j}(\mathcal{O})}^{r_j}.$$

By Step 1, c has a nonnegative progressive version in $L^1(0,T)$ almost surely. Put

$$\chi(t) = \exp\left(-\int_0^t c(s)\,\mathrm{d}s\right),$$

and define

$$\tau_m = \inf\left\{t \in [0,T] : \|y(t)\|_V + \|z(t)\|_V + \int_0^t c(s)\,\mathrm{d}s \ge m\right\} \wedge T,$$

where the empty-set infimum is $+\infty$. The V norms and the integral of c are continuous and adapted, so τ_m is a stopping time. Multiplication by χ cancels $c(t)\|e\|_H^2$. Integrating to $t \wedge \tau_m$ and taking expectations gives

$$\mathbb{E}\big[\chi(t\wedge\tau_m)\,\|e(t\wedge\tau_m)\|_H^2\big] + c_0\mathbb{E}\int_0^{t\wedge\tau_m} \chi(s)\,\|e(s)\|_V^2\,\mathrm{d}s \le 0.$$

On $[0,\tau_m)$, $0 < \chi \le 1$ and the V norms are bounded by m. By (1.4), $\|g[y] - g[z]\|_H$ is bounded by a deterministic multiple of m. Hence $2\chi(e, g[y] - g[z])_H$ is bounded and square integrable on $\Omega \times (0,T)$, and the stochastic integral has zero expectation. These bounds and $\int_0^{\tau_m} c(s)\,\mathrm{d}s \le m$ also control the other terms. For $\tau_m = 0$, all integrals and the initial difference vanish.

Step 3: Removing the stopping times. Nonnegativity of the left-hand terms and positivity of χ imply $e(t\wedge\tau_m) = 0$ almost surely for each t. Intersect these events over rational times. By continuity, the stopped paths coincide. The continuous V-valued paths are bounded on $[0,T]$ and $\int_0^T c(s)\,\mathrm{d}s < \infty$ almost surely. Hence $\tau_m \uparrow T$ and $e = 0$ on $[0,T]$ outside one null set. Only pathwise integrability of c is used, without exponential moments or H^1 time regularity. □

8. A counterexample to global null controllability

Polynomial drift growth alone does not ensure global null controllability for $y \in \mathcal{H}$ and $(u, v) \in \mathcal{U}$. We extend the test-function argument of [17] to arbitrary square-integrable adapted diffusion controls. We take $g \equiv 0$, which satisfies Assumption 1.1(ii), and a locally Lipschitz superlinear reaction.

Proposition 8.1. *Let $\mathcal{B} = B(x_*, R)$ satisfy $\overline{\mathcal{B}} \subset \mathcal{O} \setminus \overline{\mathcal{O}_0}$, and set $p = 1 + 1/n$. For each $A \in \{A_N, A_C\}$, there exists a deterministic, nonpositive initial datum $y_0 \in C_c^\infty(\mathcal{O})$, supported in $\overline{\mathcal{B}}$ and independent of T, such that, for every $T > 0$, no controls*

$$u \in L^2_{\mathbb{F}}(0, T; H_0), \qquad v \in L^2_{\mathbb{F}}(0, T; H)$$

can yield a solution $y \in \mathcal{H}$ of

$$\mathrm{d}y + Ay\,\mathrm{d}t = (-|y|^p + Bu)\,\mathrm{d}t + v\,\mathrm{d}W_t, \qquad y(0) = y_0, \tag{8.1}$$

with $y(T) = 0$ almost surely.

The conclusion also holds under the following weak formulation: for every $\psi \in C_c^\infty(\mathcal{O})$, almost surely for all $t \in [0, T]$,

$$\begin{aligned}(y(t), \psi)_H = (y_0, \psi)_H - \int_0^t (y(s), \Delta^2\psi)_H\,\mathrm{d}s - \int_0^t \int_{\mathcal{O}} |y(s,x)|^p \psi(x)\,\mathrm{d}x\,\mathrm{d}s \\ + \int_0^t (Bu(s), \psi)_H\,\mathrm{d}s + \int_0^t (v(s), \psi)_H\,\mathrm{d}W_s.\end{aligned}$$

Every variational solution with a well-defined reaction term satisfies this identity; global existence is not asserted.

Proof. Step 1: Test function. We choose the normalized nonnegative function

$$\rho(x) = c_\rho \begin{cases} \exp\bigl(-[R^2 - |x - x_*|^2]^{-1}\bigr), & x \in \mathcal{B}, \\ 0, & x \notin \mathcal{B}, \end{cases} \qquad \int_{\mathcal{O}} \rho\,\mathrm{d}x = 1.$$

Then $\rho \in C_c^\infty(\mathcal{O}) \cap D(A)$, $B^*\rho = 0$, and $A\rho = \Delta^2\rho$ for both realizations. Hence $a(y, \rho) = (y, A\rho)_H$ for $y \in V$, so this test is admissible in the weak formulation.

Let $p' = p/(p-1)$. Applying Young's inequality to $\rho^{1/p}|r|$ and $\rho^{-1/p}|A\rho|$ gives

$$|r|\,|A\rho| \le \tfrac{1}{2}\rho|r|^p + C_p|A\rho|^{p'}\rho^{-1/(p-1)} \qquad (\rho > 0). \tag{8.2}$$

The remainder is integrable:

$$C_\rho := C_p \int_{\mathcal{B}} |A\rho|^{p'}\rho^{-1/(p-1)}\,\mathrm{d}x < \infty.$$

Indeed, set $s = R^2 - |x - x_*|^2$ in $\mathcal{B}$. This polynomial has bounded derivatives, and each differentiation introduces at most two inverse powers of s. Hence

$$|A\rho| \le C\rho(1 + s^{-8}).$$

Since $p' - 1/(p-1) = 1$, it follows that

$$|A\rho|^{p'}\rho^{-1/(p-1)} \le C\rho(1 + s^{-8p'}).$$

The product $e^{-1/s}s^{-8p'}$ is bounded and extends by zero at $s = 0$, proving integrability. On $\mathcal{O} \setminus \mathcal{B}$, ρ and $A\rho$ vanish; set the remainder in (8.2) to zero there.

Step 2: Expectation identity. Suppose (u, v) yields a solution $y \in \mathcal{H}$ on $[0, T]$. Since $1 < p = 1 + 1/n \leq 2$, Hölder's inequality on $\Omega \times (0, T) \times \mathcal{O}$, whose measure is $T|\mathcal{O}|$, gives integrability of the reaction pairing with ρ in $L^1(\Omega \times (0, T))$. The weak formulation and Fubini's theorem are therefore justified without an $L^2(0, T; H)$ bound on the reaction. The linear pairing is also integrable. Since v is admissible, the projected stochastic integral is a square-integrable martingale with zero expectation at every deterministic time, even when v is state-dependent.

Set $m(t) = -\mathbb{E}(y(t), \rho)_H$. Testing with ρ eliminates the control u since $B^*\rho = 0$. Taking expectations gives, for every $t \in [0, T]$,

$$m(t) = m(0) + \int_0^t \left[\mathbb{E}(y(s), A\rho)_H + \mathbb{E}\int_{\mathcal{O}} \rho|y(s)|^p \,\mathrm{d}x\right] \mathrm{d}s.$$

Thus m is absolutely continuous.

Step 3: Moment inequality. Differentiate almost everywhere and apply (8.2) to obtain

$$m'(t) = \mathbb{E}(y, A\rho)_H + \mathbb{E}\int_{\mathcal{O}} \rho|y|^p \,\mathrm{d}x \geq \tfrac{1}{2}\mathbb{E}\int_{\mathcal{O}} \rho|y|^p \,\mathrm{d}x - C_\rho \geq \tfrac{1}{2}|m(t)|^p - C_\rho. \tag{8.3}$$

Integrating Young's inequality over space and probability gives the first inequality. Jensen's inequality for the probability measure $\mathrm{d}\mathbb{P}(\omega)\rho(x)\,\mathrm{d}x$, with $\rho \geq 0$ and $\int_{\mathcal{O}} \rho(x)\,\mathrm{d}x = 1$, gives the lower bound $|m(t)|^p$. No sign preservation or maximum principle is used.

Step 4: Contradiction. Choose $K > (2C_\rho)^{1/p}$ and $y_0 = -K\rho/\|\rho\|_H^2$, a deterministic nonpositive element of $C_c^\infty(\mathcal{O})$ supported in $\overline{\mathcal{B}}$, with $m(0) = K$. Set $f(r) = \frac{1}{2}|r|^p - C_\rho$, so $f(K) > 0$. Since m is continuous on $[0, T]$ and f is locally Lipschitz, positive-part comparison applies. The chain rule for $q = (K - m)_+$, (8.3), and Gronwall's inequality with $q(0) = 0$ give $q \equiv 0$. Hence $m(t) \geq K > 0$ on $[0, T]$. This contradicts $y(T) = 0$ almost surely, which implies $m(T) = 0$. The contradiction holds for every admissible solution on $[0, T]$; C_ρ, K and y_0 are independent of T and the controls. □

Remark (An upper bound on admissible existence times)**.** For any solution with this datum, Step 4 gives $m \geq K$. Thus (8.3) implies $m' \geq a_K m^p$, with $a_K = \frac{1}{2} - C_\rho/K^p > 0$. Since m is absolutely continuous and $m \geq K > 0$, so is m^{1-p}. Multiplication by $(1 - p)m^{-p}$ reverses the inequality: $(m^{1-p})' \leq -(p - 1)a_K$ almost everywhere. Integrating over $(0, t)$ gives

$$m(t)^{1-p} \leq K^{1-p} - (p - 1)a_K t, \qquad T_* := \frac{K^{1-p}}{(p - 1)(\frac{1}{2} - C_\rho/K^p)}.$$

At $t = T_*$, the right-hand side is zero, whereas the left-hand side is positive for finite $m(T_*) \geq K$. Hence no energy-class solution exists on $[0, T]$ for $T \geq T_*$, uniformly over admissible controls. On shorter intervals, any solution has a positive moment and therefore cannot have a zero terminal state.

Corollary 8.2. *Under the geometric assumption of Proposition 8.1, let g satisfy Assumption 1.1(ii). The conclusion of that proposition remains valid when the diffusion in (8.1) is replaced by $g[y]+v$. The initial datum can be chosen independently of g.*

Proof. Suppose, to the contrary, that controls $(u,v)\in\mathcal{U}$ yield a null trajectory $y\in\mathcal{H}$ for the modified diffusion. The diffusion assumption and the definition of $\mathcal{H}$ give

$$\mathbb{E}\int_0^T \|g[y]\|_H^2\,\mathrm{d}t \le 2a_g\mathbb{E}\int_0^T \|y\|_V^2\,\mathrm{d}t + 2b_g\mathbb{E}\int_0^T \|y\|_H^2\,\mathrm{d}t + 2\|g_*\|^2_{L^2_{\mathbb{F}}(0,T;H)} < \infty.$$

By joint measurability, $g[y]$ is progressive. Hence $\widetilde{v}=g[y]+v$ belongs to $L^2_{\mathbb{F}}(0,T;H)$. The state y with controls $(u,\widetilde{v})$ satisfies (8.1) and its weak formulation, contradicting Proposition 8.1. The same datum works for all g satisfying Assumption 1.1(ii). Without a terminal condition, the same reduction gives the existence-time bound T_*. □

The deterministic reaction $f_0(r)=-|r|^p$, $p=1+1/n$, satisfies $f_0(0)=0$ and the global increment bound

$$|f_0(r)-f_0(s)| \le p\big(|r|^{p-1}+|s|^{p-1}\big)|r-s|.$$

For $n\le 4$, p_0^* is infinite; for $n>4$, $1+1/n\le (n+4)/(n-4)=p_0^*$. Thus the reaction satisfies Assumption 1.13 with $J=\{0\}$ and is not globally Lipschitz. The chosen initial moment exceeds the threshold; Theorem 1.14 applies to small data. The sign structure of $-|y|^{p-1}y$ and the divergence structure of $\Delta(y^3-y)$ do not yield (8.3), so this argument gives no obstruction for either nonlinearity.

Appendix A. The strict diffusion coercivity condition

We explain the strict condition $a_g<2$ in Assumption 1.1; see Remark 1.2. Consider the uncontrolled Navier equation

$$\mathrm{d}Y+\Delta^2 Y\,\mathrm{d}t=\sigma\Delta Y\,\mathrm{d}W_t,\qquad \sigma\in\mathbb{R}.$$

Here $g(Y)=\sigma\Delta Y$ gives $\|g(Y)-g(Z)\|_H^2=\sigma^2\|Y-Z\|_{V_N}^2$, so one can take $a_g=\sigma^2$.

Let (e_k) be an orthonormal Dirichlet Laplacian eigenbasis, $-\Delta e_k=\lambda_k e_k$, and take deterministic $Y_0\in H$. Set $a_k=(Y_0,e_k)_H$. A variational solution $Y\in\mathcal{H}$ would have coordinates $Y_k=(Y,e_k)_H$ satisfying

$$\mathrm{d}Y_k=-\lambda_k^2 Y_k\,\mathrm{d}t-\sigma\lambda_k Y_k\,\mathrm{d}W_t,\qquad Y_k(0)=a_k.$$

Uniqueness for the scalar equations gives

$$Y_k(t)=a_k\exp\left[-\left(1+\frac{\sigma^2}{2}\right)\lambda_k^2 t-\sigma\lambda_k W_t\right],$$

and hence $\mathbb{E}|Y_k(t)|^2=|a_k|^2e^{(\sigma^2-2)\lambda_k^2 t}$.

For $c:=\sigma^2-2>0$, choose (k_j) with $\lambda_{k_j}^2\ge j^2$ and set $a_{k_j}=e^{-cT\lambda_{k_j}^2/2}$, all other coefficients being zero. Then $Y_0\in H$. By Parseval's identity and monotone convergence, $\mathbb{E}\|Y(T)\|_H^2=\sum_j\mathbb{E}|Y_{k_j}(T)|^2=\sum_j 1=\infty$, contradicting $Y\in\mathcal{H}$.

For $\sigma^2 = 2$, take any $Y_0 \in H \setminus V_N$. The spectral characterization of $V_N = D(A_N^{1/2})$ gives $\sum_k \lambda_k^2 |a_k|^2 = \infty$. The moment formula and Tonelli's theorem yield

$$\mathbb{E} \int_0^T \|Y(t)\|_{V_N}^2 \, \mathrm{d}t = T \sum_k \lambda_k^2 |a_k|^2 = \infty.$$

Again no solution lies in $\mathcal{H}$. Thus, for well-posedness with every prescribed control pair, $a_g < 2$ cannot generally be weakened to $a_g \leq 2$. Compensation through the diffusion control remains possible.

Appendix B. Well-posedness for prescribed controls

We prove Proposition 2.2. The constant $C_{\mathcal{O}}$ depends on the spatial embeddings for the chosen realization and may vary from line to line.

Step 1: Linear solution map. Consider

$$\mathrm{d}Y + AY \, \mathrm{d}t = f \, \mathrm{d}t + G \, \mathrm{d}W_t, \qquad Y(0) = Y_0,$$

with $f, G \in L^2_{\mathbb{F}}(0, T; H)$ and $Y_0 \in L^2(\Omega, \mathcal{F}_0; H)$.

Let P_N project orthogonally onto the first N eigenfunctions of A. The projected data $(P_N Y_0, P_N f, P_N G)$ give a continuous adapted solution $Y_N \in S^2(H)$.

Here $S^2(H) = L^2_{\mathbb{F}}(\Omega; C([0, T]; H))$. By localized Itô's formula and the BDG and Cauchy–Schwarz inequalities, we obtain an energy estimate uniform in N. Its constant depends only on the domain, final time and boundary realization. Since $P_N \to I$ strongly on H, the data converge in L^2. The difference estimate for $Y_N - Y_M$ shows that (Y_N) is Cauchy in $\mathcal{H}$.

Using the bounded extension $A : V \to V'$ and the BDG inequality, we pass to the variational and stochastic limits, respectively. The difference estimate yields uniqueness and a linear solution map; see [28].

Step 2: Energy-space contraction. The spatial embeddings, Assumption 1.1(i), and Fubini's theorem give, outside a common base null set independent of $z, w \in V$,

$$\|F[z] - F[w]\|_H \leq L_V \|z - w\|_V, \qquad L_V \leq C_{\mathcal{O}}(L_0^2 + L_1^2 + L_2^2)^{1/2}.$$

Since $F[0] = 0$, Lemma 2.1 applies to F and g. Their compositions with progressive V-valued processes are therefore progressive and square integrable.

The gradient satisfies the interpolation estimate

$$\|\nabla z\|_H^2 = -(z, \Delta z)_H \leq \|z\|_H \|z\|_V, \qquad z \in V. \tag{B.1}$$

For $X \in L^2_{\mathbb{F}}(0, T; V)$, let $\mathscr{T}X$ be the solution obtained in Step 1 with initial datum y_0, drift $F[X] + Bu$ and diffusion $g[X] + v$. Then $\mathscr{T}X \in \mathcal{H}$. Equip $L^2_{\mathbb{F}}(0, T; V)$ with the equivalent complete norm

$$\|X\|_\beta^2 = \mathbb{E} \int_0^T e^{-\beta t} \left(\frac{\beta}{2} \|X(t)\|_H^2 + 2\|X(t)\|_V^2 \right) \mathrm{d}t, \qquad \beta > 0.$$

Set $Y = \mathscr{T}X - \mathscr{T}\widetilde{X}$ and $e = X - \widetilde{X}$, so $Y(0) = 0$. Apply Itô's formula to $e^{-\beta t}\|Y(t)\|_H^2$ and take expectations. Young's inequality and the coefficient bounds give

$$\|\mathscr{T}X - \mathscr{T}\widetilde{X}\|_\beta^2 \le q_\beta \|X - \widetilde{X}\|_\beta^2, \qquad q_\beta = \max\left\{\frac{a_g}{2} + \frac{L_V^2}{\beta}, \frac{2b_g}{\beta}\right\}.$$

Set $\eta = 2 - a_g > 0$ and choose

$$\beta = 1 + 4b_g + \frac{4L_V^2}{\eta}.$$

Since $0 < \eta \le 2$, we have $q_\beta \le 1 - \eta/4 < 1$ and hence a unique fixed point $X \in L^2_{\mathbb{F}}(0,T;V)$. Set $y = \mathscr{T}X \in \mathcal{H}$. Because $y = X$ in $L^2_{\mathbb{F}}(0,T;V)$, Lemma 2.1 gives $F[y] = F[X]$ and $g[y] = g[X]$. This proves (2.3). By linear uniqueness, every variational solution is a fixed point of $\mathscr{T}$; continuity in H then gives indistinguishability.

Step 3: Stability. For y_1, y_2 with different data and controls but the same coefficients, set $e = y_1 - y_2$, $\delta F = F[y_1] - F[y_2]$, and $h = g[y_1] - g[y_2] + \delta v$. With $\epsilon = \eta/[2(1 + a_g)]$, the coefficient bounds give

$$\begin{aligned} \|h\|_H^2 &\le (2 - \eta/2)\|e\|_V^2 + (1 + \epsilon)b_g\|e\|_H^2 + (1 + \epsilon^{-1})\|\delta v\|_H^2, \\ 2|(e, \delta F + B\delta u)_H| &\le \frac{\eta}{4}\|e\|_V^2 + (1 + C_\eta L_V^2)\|e\|_H^2 + \|\delta u\|_{H_0}^2, \end{aligned} \tag{B.2}$$

where $C_\eta = 4/\eta$. Since $(1 + \epsilon)a_g \le 2 - \eta/2$, the preceding estimates leave a coercivity margin of $\eta/4$. Taking expectations in the Itô formula for $\|e(t)\|_H^2$, after localization if necessary, gives

$$\mathbb{E}\|e(t)\|_H^2 + \frac{\eta}{4}\mathbb{E}\int_0^t \|e\|_V^2 \,\mathrm{d}\tau \le R_t + K_E \int_0^t \mathbb{E}\|e\|_H^2 \,\mathrm{d}\tau, \tag{B.3}$$

where

$$R_t = \mathbb{E}\|e(0)\|_H^2 + \mathbb{E}\int_0^t \left(\|\delta u\|_{H_0}^2 + (1 + \epsilon^{-1})\|\delta v\|_H^2\right) \mathrm{d}\tau, \qquad K_E = 1 + (1 + \epsilon)b_g + C_\eta L_V^2.$$

Gronwall's inequality bounds $\mathbb{E}\|e(t)\|_H^2$ uniformly in t; substitution into (B.3) controls the integrated V-norm:

$$\sup_{t \le T}\mathbb{E}\|e(t)\|_H^2 + \mathbb{E}\int_0^T \|e(t)\|_V^2 \,\mathrm{d}t \le C_E\big(\mathbb{E}\|e(0)\|_H^2 + \|\delta u\|^2_{L^2_{\mathbb{F}}(0,T;H_0)} + \|\delta v\|^2_{L^2_{\mathbb{F}}(0,T;H)}\big). \tag{B.4}$$

By the BDG and Young inequalities, the martingale supremum is bounded by one quarter of the state supremum plus diffusion energy. We control the latter by (B.2) and (B.4). We take the supremum in the Itô formula and absorb the state-supremum term to obtain (2.4). The coercivity margin is therefore retained: $a_g < 2$ suffices, with the stated constant dependence. The term g_* cancels in the difference equation.

Step 4: A priori and anisotropic bounds. Since $F[0] = 0$, the zero state solves (2.3) with zero initial datum and controls $(0, -g_*)$. Comparing y with this solution in (2.4) and applying the squared triangle inequality proves (2.5) after increasing C_E.

Finally, let $e = z - w$, $x = \|e\|_H$, and $d = \|e\|_V$. By (B.1) and the elliptic estimate $\|D^2 e\|_{L^2(\mathcal{O})} \le C_{\mathcal{O},2} d$, for every $\varepsilon > 0$ we have

$$\begin{aligned} 2|(e, F[z] - F[w])_H| &\le 2L_0 x^2 + 2L_1 x^{3/2} d^{1/2} + 2C_{\mathcal{O},2} L_2 x d \\ &\le \varepsilon d^2 + C_{\mathcal{O}}\big(L_0 + \varepsilon^{-1/3} L_1^{4/3} + \varepsilon^{-1} L_2^2\big) x^2. \end{aligned} \tag{B.5}$$

The second inequality follows from Young's inequality.

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